

DFlowNovo: Continuous-Time Discrete Flow Matching and Dynamic Knapsack Guidance for High-Throughput De Novo Peptide Sequencing

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Abstract

Tandem mass spectrometry (MS/MS) enables high-throughput identification of proteins, yet identifying novel peptides without reference databases—*de novo* peptide sequencing—remains computationally challenging. Existing state-of-the-art approaches rely on autoregressive transformers that predict amino acid sequences token by token in $\mathcal{O}(L)$ sequential steps. While accurate, autoregressive decoding suffers from error propagation, directional fragmentation asymmetry between complementary *b*- and *y*-ions, and severe latency bottlenecks. Continuous normalizing flows alleviate sequential decoding but suffer from severe discretization error upon rounding continuous latent states to discrete amino acid masses.

In this work, we introduce **DFlowNovo**, a non-autoregressive generative framework that models *de novo* peptide sequencing as continuous-time Discrete Flow Matching (DFM) over the probability simplex of amino acids, coupled with exact Dynamic Programming Knapsack reachability filtering (KnapsackDP). DFlowNovo generates complete peptide sequences in a constant budget of $K = 20$ reverse flow steps, achieving an inference throughput of 227.1–255.8 spectra per second on an NVIDIA GPU (a $4.3\times$ to $4.9\times$ speedup over autoregressive beam search). On the standardized 104k Nine-Species test benchmark, DFlowNovo achieves 68.28% strict exact match and 83.01% amino acid F1 score, outperforming InstaNovo v1.2 (65.48%) and accepting 86,412 peptide-spectrum matches at 80% precision. Furthermore, we dissect the physical foundations of mass spectrometry, demonstrating that remaining prediction discrepancies on human data are predominantly driven by isomeric Leucine/Isoleucine equivalence (113.084 Da) which requires radical-driven *w*-ion fragmentation to break.

Résumé

La spectrométrie de masse en tandem (MS/MS) permet l'identification des protéines à haut débit, mais le séquençage *de novo*—l'identification des peptides sans base de données de référence—reste un défi informatique majeur. Les modèles autorégressifs actuels prédisent les acides aminés de manière séquentielle en $\mathcal{O}(L)$ étapes, ce qui induit une propagation des erreurs, une asymétrie entre ions complémentaires *b* et *y*, et une forte latence. Les modèles à flux continu évitent ce coût séquentiel mais souffrent d'une perte de discrétisation critique lors de la projection sur les masses discrètes des résidus.

Ce travail présente **DFlowNovo**, un cadre génératif non-autorégressif formulant le séquençage *de novo* comme un appariement de flux discret (DFM) en temps continu sur le simplexe de probabilité, guidé par une programmation dynamique exacte du sac à dos (KnapsackDP). DFlowNovo décode les séquences peptidiques entières en un nombre fixe de 20 pas d'intégration, atteignant une vitesse de 227 à 256 spectres par seconde (4,3 à 4,9 fois plus rapide que les modèles autorégressifs). Sur le banc d'essai biologique des Neuf Espèces (104 163 spectres), DFlowNovo atteint 68,28% d'exactitude stricte et 83,01% de score F1, surpassant InstaNovo v1.2 tout en identifiant 86 412 spectres à 80% de précision. Enfin, une analyse biophysique démontre que les ambiguïtés résiduelles découlent de l'isomérisme fondamentale Leucine/Isoleucine, indissociable par fragmentation collisionnelle standard.

Resource Availability: The complete source code, trained model checkpoints, interactive demonstration suites, and benchmark data manifests are provided in Appendix D.

Declaration

I, the undersigned, hereby declare that the work contained in this research project is my original work, and that any work done by others or by myself previously has been acknowledged and referenced accordingly.

Scan your signature

Joël Gédéon, September 2026

Contents

Abstract	i
1 Introduction to De Novo Peptide Sequencing and Mass Spectrometry	1
1.1 The Proteomics Frontier and Tandem Mass Spectrometry	1
1.2 Database Search Engines versus De Novo Sequencing	1
1.3 Physical Principles of Collisional Fragmentation	2
1.4 Computational Complexity and Predecessor Algorithms	3
1.5 Outline of Contributions	4
2 Mathematical Foundations of Continuous-Time Discrete Flow Matching	5
2.1 The Discrete Sequence Simplex and State Space	5
2.2 Continuous-Time Markov Chains and the Kolmogorov Equation	5
2.3 Generative and Data-Conditional Probability Flows	7
2.4 Closed-Form Conditional Rate Matrices and Jump Minimality	7
2.5 Detailed Balance and Tunable CTMC Stochasticity	8
2.6 Probability Interpolants, Jump Flux, and Schedule Dynamics	9
2.7 Discrete Flow Matching Objective and Training Algorithm	11
2.8 Reverse-Time CTMC Discrete Flow Sampling	11
3 The DFlowNovo Framework: Architecture and Dynamic Knapsack Guidance	14
3.1 Part A: Neural Network Model Architecture (Training and Forward Pass)	14
3.2 Part B: Mass-Guided Iterative Sampling and Dynamic Filtering Pipeline (Inference)	18
4 Empirical Benchmark Evaluations and Quantitative Comparisons	23
4.1 Benchmark Datasets and Standardized Evaluation Protocol	23
4.2 Evaluation Metrics and Biophysical Importance	23
4.3 Competitive Baseline Models	25
4.4 Empirical Benchmark Results	26
4.5 Precision-Coverage Dynamics and Statistical Significance	26
4.6 Inference Throughput and Latency Scaling	28
5 Mass Spectrometry Physics, Biological Fidelity and Error Dissection	29
5.1 The Isobaric Leucine/Isoleucine Ambiguity and Radical Ions	29
5.2 The Proline Effect and Asymmetric Backbone Cleavage	30
5.3 Long Peptide Dynamics and Length-Weighted Representation Balancing	31
5.4 Mechanistic Case Studies: Four Canonical Physical Regimes	32
6 Conclusions and Future Perspectives	34
6.1 Summary of Contributions	34
6.2 Methodological Limitations	34
6.3 Future Research Directions	35
A Detailed Benchmark Results and Per-Species Stratification	36
A.1 Per-Species Benchmark Stratification	36
A.2 Cross-Species Generalization and Proteomic Complexity	36

A.3	Extended Multi-Model Comparison on Standard Benchmarks	37
A.4	Comprehensive Catalog of De Novo Prediction Case Studies	38
B	Hyperparameter Configurations and Training Protocols	42
B.1	Model Architecture Specifications	42
B.2	Optimization and Training Dynamics	43
B.3	Inference and Dynamic Knapsack Guidance Settings	43
B.4	Empirical Training and Validation Trajectories	43
B.5	Empirical CTMC Flow Dynamics and Schedule Analysis	44
C	Theoretical Proofs of Discrete Flow Matching Theorems	47
C.1	Proof of Proposition 2.3.1: Marginal Rate Matrix Expectation	47
C.2	Proof of Proposition 2.4.1: Verification of Canonical Rate Matrix	48
C.3	Proof of Proposition 2.5.1: Detailed Balance Invariance	49
C.4	Multi-Trajectory Empirical Flow Dynamics and Positional Entropy Collapse	49
D	Software and Resource Availability	55
D.1	Open-Source Codebase and Repository Architecture	55
D.2	Pre-Trained Model Weights and Checkpoints	56
D.3	Interactive Cloud Demonstration Studio	56
D.4	Standardized Benchmark Proteomics Datasets	56
D.5	Computational Infrastructure and Hardware Environment	57
	References	61

1. Introduction to De Novo Peptide Sequencing and Mass Spectrometry

Proteins are the primary functional macromolecular effectors of biological processes, executing enzymatic catalysis, structural scaffolding, signal transduction, and immunological defense within all living organisms. Deciphering the exact amino acid sequence of proteins and peptides is therefore foundational to molecular biology, pharmacology, clinical oncology, and biotechnology. Tandem mass spectrometry (MS/MS) coupled with liquid chromatography (LC-MS/MS) has emerged as the premier high-throughput analytical technology for resolving complex cellular proteomes. However, translating high-dimensional, noisy fragmentation spectra into accurate primary sequences remains one of the most demanding inverse problems in computational biology.

1.1 The Proteomics Frontier and Tandem Mass Spectrometry

In a standard bottom-up proteomics workflow, extracted proteins are enzymatically digested into shorter peptide fragments (typically 7 to 35 amino acids in length) using endopeptidases such as trypsin (Olsen et al., 2004; Vandermarliere et al., 2013). These peptides are separated by reversed-phase high-performance liquid chromatography and introduced into a mass spectrometer via electrospray ionization (ESI).

During the initial survey stage (MS1), the instrument measures the mass-to-charge ratios (m/z) and isotopic abundance envelopes of intact, multiply protonated peptide precursor ions. Subsequently, selected precursor ions within narrow isolation windows are accelerated into a collision cell for tandem mass spectrometry (MS2). In Higher-Energy Collisional Dissociation (HCD) (Olsen et al., 2007), precursor ions collide with neutral inert gas molecules (such as nitrogen or argon), converting kinetic energy into vibrational excitation. This vibrational excitation induces unimolecular thermal decomposition across the weakest covalent bonds of the peptide backbone, predominantly the amide peptide bonds (Paizs and Suhai, 2005). The resulting fragment ions are injected into an ultra-high-resolution mass analyzer, such as an Orbitrap (Makarov, 2000; Zubarev and Mann, 2007), which records their exact m/z and intensity values, generating an experimental MS2 fragmentation spectrum.

1.2 Database Search Engines versus De Novo Sequencing

For over two decades, the dominant paradigm for analyzing tandem mass spectra has been database-dependent search algorithms, pioneered by SEQUEST (Eng et al., 1994) and subsequently refined in tools such as Mascot (Perkins et al., 1999), MaxQuant (Cox and Mann, 2008), and MSFragger (Kong et al., 2017), where spectrum-to-sequence matches are statistically validated using target-decoy false discovery rate (FDR) estimation (Elias and Gygi, 2007). Database search engines operate by matching experimental spectra against *in silico* theoretical spectra generated from reference proteome sequence databases (such as UniProt). While highly sensitive and computationally tractable for well-characterized model organisms, database searching possesses severe intrinsic limitations (Nesvizhskii, 2014):

- (i) **Unannotated Genomes and Non-Model Species:** For the vast majority of biological species on Earth, high-quality reference genomes or proteome annotations do not exist. Database searching fails entirely when the target organism is unsequenced (Nesvizhskii, 2014; Aebersold and Mann, 2016).

- (ii) **Cancer Neoantigens and Somatic Mutations:** Malignant cells generate novel peptide sequences resulting from point mutations, gene fusions, alternative splicing, and frame-shift events that are absent from canonical reference databases. Identifying these tumor-specific antigens is vital for personalized immunotherapy and cancer vaccine development (Schumacher and Schreiber, 2015; Bassani-Sternberg et al., 2016).
- (iii) **Immunopeptidomics and Antibody Repertoires:** B-cell and T-cell receptor repertoires undergo somatic hypermutation and $V(D)J$ genetic recombination, yielding hypervariable antigen-binding complementarity-determining regions (CDRs). Assembling monoclonal antibodies or polyclonal serum repertoires directly from biological fluids requires sequence identification without genomic references (Purcell et al., 2019; Bonissone and Pevzner, 2016).
- (iv) **Post-Translational Modifications (PTMs):** Variable chemical modifications (phosphorylation, acetylation, ubiquitination, methylation) expand the combinatorial search space exponentially, rendering brute-force database searches computationally intractable (Chick et al., 2015; Kong et al., 2017).

To overcome these barriers, *de novo* peptide sequencing aims to reconstruct the exact primary amino acid sequence directly from the experimental spectrum S and precursor mass M_{prec} , relying solely on the physical laws of mass conservation and fragmentation chemistry.

1.3 Physical Principles of Collisional Fragmentation

To understand the mathematical structure of the sequencing problem, we examine the nomenclature and mechanisms governing peptide fragmentation. Following the standardized Roepstorff-Biemann nomenclature (Roepstorff and Fohlman, 1984; Biemann, 1988), cleavage of the amide bond ($-CO-NH-$) along the peptide backbone generates two complementary series of fragment ions:

1.3.1 Definition (Prefix and Suffix Fragment Ions). Let $Y = (y_1, y_2, \dots, y_L)$ denote a peptide of length L , where each $y_i \in \mathcal{V}_{\text{AA}}$ is an amino acid residue with monoisotopic mass $m(y_i)$.

- (i) A ***b*-ion** of length $k \in \{1, \dots, L-1\}$ contains the N-terminal prefix sequence $(y_1, \dots, y_k)$. Its theoretical monoisotopic mass is:

$$m(b_k) = \sum_{j=1}^k m(y_j) + m(\text{H}^+) \quad (1.3.1)$$

where $m(\text{H}^+) \approx 1.007276$ Da is the proton mass.

- (ii) A ***y*-ion** of length $k \in \{1, \dots, L-1\}$ contains the C-terminal suffix sequence $(y_{L-k+1}, \dots, y_L)$. Its theoretical monoisotopic mass is:

$$m(y_k) = \sum_{j=L-k+1}^L m(y_j) + m(\text{H}_2\text{O}) + m(\text{H}^+) \quad (1.3.2)$$

where $m(\text{H}_2\text{O}) \approx 18.010565$ Da represents the terminal water molecule retained by the carboxylic acid terminus.

Because cleavage of a specific peptide bond produces either a *b*-ion or a *y*-ion, every b_k ion has a complementary y_{L-k} ion satisfying the fundamental mass conservation duality:

$$m(b_k) + m(y_{L-k}) = M_{\text{prec}} + m(\text{H}^+) \quad (1.3.3)$$

where $M_{\text{prec}} = \sum_{i=1}^L m(y_i) + m(\text{H}_2\text{O}) + m(\text{H}^+)$ is the singly protonated intact precursor mass. The mass difference between two consecutive fragment ions of the same series reveals the mass of the intervening amino acid:

$$m(b_k) - m(b_{k-1}) = m(y_{L-k+1}) - m(y_{L-k}) = m(y_k) \quad (1.3.4)$$

In an idealized, noise-free mass spectrum, a complete ladder of b -ions or y -ions would permit trivial sequence extraction by reading off adjacent mass intervals. However, real experimental spectra deviate substantially from this idealization due to missing fragmentation peaks, internal fragments, neutral losses (H_2O , NH_3), multiply charged ions, chemical noise, and instrument measurement jitter (Paizs and Suhai, 2005).

1.4 Computational Complexity and Predecessor Algorithms

The primary amino acid alphabet consists of 20 canonical residues with monoisotopic masses ranging from Glycine (57.021 Da) to Tryptophan (186.079 Da). For a typical tryptic peptide of length $L \approx 15$, the number of possible sequence permutations is $20^{15} \approx 3.28 \times 10^{19}$. Filtering by precursor neutral mass restricts the candidate space, but still leaves millions of isobaric sequence permutations.

Early computational approaches framed *de novo* sequencing as finding the highest-scoring path through a directed acyclic spectrum graph (Dancik et al., 1999; Chen et al., 2001; Frank and Pevzner, 2005). Nodes represented observed experimental peaks, and directed edges corresponded to valid amino acid residue masses. While dynamic programming could compute optimal paths in polynomial time, these heuristic graph methods struggled with missing peaks, false ladder connections, and non-linear peak intensity variations.

Recent advances in deep learning have significantly improved *de novo* sequencing performance. DeepNovo (Tran et al., 2017) combined convolutional neural networks (CNNs) for peak feature extraction with long short-term memory (LSTM) networks and beam search decoding. PointNovo (Qiao et al., 2021) introduced continuous order-invariant spectrum encodings. Casanovo (Yilmaz et al., 2022) adapted the Transformer architecture, framing sequencing as an autoregressive machine translation problem from peak coordinates to amino acid tokens. More recently, InstaNovo (Eloff et al., 2025) incorporated mass-guided beam search with deep autoregressive Transformers, setting a new benchmark for sequencing accuracy across large-scale human proteomes.

Despite their success, existing autoregressive models suffer from fundamental algorithmic limitations:

- (i) **Sequential Decoding Latency:** Generating a sequence of length L requires L sequential forward passes through the Transformer decoder. With typical peptide lengths of $L \in [7, 30]$ and beam sizes of $B \in [5, 10]$, decoding speed is capped at 25 to 55 spectra per second on modern GPUs, creating an acute bottleneck for clinical high-throughput repositories containing tens of millions of spectra.
- (ii) **Error Propagation and Exposure Bias:** Because autoregressive models generate tokens left-to-right, an error made on an early residue propagates through the entire remaining sequence, leading to sequence-level failure.
- (iii) **Directional Ion Asymmetry:** Left-to-right generation favors b -ions over y -ions, violating the physical bilateral symmetry of MS/MS fragmentation expressed in Equation 1.3.3.
- (iv) **Discretization Failure in Continuous Flows:** Continuous normalizing flow approaches, such as PowerNovo2 (Petrovskiy et al., 2026), model continuous latent vectors rather than discrete

sequences. Projecting continuous vectors onto discrete amino acid masses causes massive discretization error, resulting in low exact match rates (3.16% on standard biological benchmarks).

1.5 Outline of Contributions

To simultaneously overcome the latency bottleneck of autoregressive models and the discretization failure of continuous normalizing flows, this thesis introduces **DFlowNovo**, a continuous-time Discrete Flow Matching framework tailored for high-throughput *de novo* peptide sequencing.

The primary contributions of this thesis are:

- (1) **Discrete Flow Formulation for Proteomics:** We establish the first continuous-time Discrete Flow Matching (DFM) formulation directly defined over the discrete probability simplex of amino acid residues, avoiding continuous discretization errors entirely.
- (2) **Exact Polynomial-Time Knapsack Guidance (KnapsackDP):** We derive and implement a GPU-accelerated dynamic programming reachability filter that prunes mathematically invalid amino acid transitions during flow integration in $\mathcal{O}(1)$ time per token, enforcing 100% mass compliance.
- (3) **Empirical State-of-the-Art Accuracy:** Across the standardized 104k Nine-Species biological benchmark (Tran et al., 2017) (published on the Hugging Face Hub as `InstaDeepAI/ms_ninespecies_benchmark`) DFlowNovo achieves 68.28% strict exact match and 83.01% residue F1 score, outperforming InstaNovo v1.2 (65.48%) and accepting 86,412 peptide-spectrum matches at 80% precision.
- (4) **High-Throughput Non-Autoregressive Inference:** DFlowNovo decodes all residue positions in parallel within a fixed budget of $K = 20$ integration steps, achieving 227.1 to 255.8 spectra per second on an NVIDIA GPU—a $4.3\times$ to $4.9\times$ throughput increase over autoregressive beam search.
- (5) **Physical Dissection of Fragmentation Chemistry:** We analyze the biological and mass spectrometry limits of sequencing, proving that 30.88% of remaining errors on human benchmarks originate from the fundamental isobaric equivalence of Leucine and Isoleucine (113.084 Da), which cannot be resolved without radical-driven *w*-ion dissociation.

The remainder of this thesis is structured as follows. Chapter 2 details the mathematical foundations of continuous-time discrete flow matching and Continuous-Time Markov Chains. Chapter 3 presents the architectural design of DFlowNovo and the exact dynamic programming knapsack reachability filter. Chapter 4 reports extensive empirical benchmarks, latency comparisons, and ablation experiments. Chapter 5 provides an in-depth physical and chemical error analysis. Finally, Chapter 6 summarizes the findings and outlines promising future directions.

2. Mathematical Foundations of Continuous-Time Discrete Flow Matching

Generative modeling of discrete sequences has historically relied on autoregressive factorizations or discrete diffusion models based on discrete-time Markov chains. While continuous normalizing flows (Chen et al., 2018; Lipman et al., 2023) define invertible vector fields and probability density paths on smooth Euclidean or Riemannian manifolds, extending them to discrete categorical domains via continuous relaxations (Stark et al., 2024) can introduce boundary mismatch, geometric distortion, and quantization errors.

This chapter presents the mathematical framework of continuous-time Discrete Flow Models (DFMs) developed by Campbell et al. (2024) and Gat et al. (2024). By formulating probability transport directly over the discrete probability simplex via time-inhomogeneous Continuous-Time Markov Chains (CTMCs), DFMs construct simulation-free generative models that operate natively on discrete configuration spaces.

2.1 The Discrete Sequence Simplex and State Space

Let $\mathcal{X} = \{1, 2, \dots, S\}$ denote a finite vocabulary of S discrete categorical states, containing an absorbing mask state denoted by $[M] \in \mathcal{X}$. A discrete sequence of fixed dimension D is an element of the Cartesian product state space:

$$\mathcal{X}^D = \underbrace{\mathcal{X} \times \mathcal{X} \times \dots \times \mathcal{X}}_{D \text{ coordinates}} \quad (2.1.1)$$

The cardinality of this configuration space is $|\mathcal{X}^D| = S^D$. A joint probability distribution over $\mathcal{X}^D$ is represented as a probability mass vector $p \in \Delta^{S^D-1}$ residing on the probability simplex:

$$\Delta^{S^D-1} = \left\{ p \in \mathbb{R}^{S^D} \left| \sum_{x \in \mathcal{X}^D} p(x) = 1, \quad p(x) \geq 0 \quad \forall x \in \mathcal{X}^D \right. \right\} \quad (2.1.2)$$

For any individual coordinate $d \in \{1, \dots, D\}$, the marginal distribution p^d resides on the $(S-1)$ -dimensional categorical simplex Δ^{S-1} .

Because $|\mathcal{X}^D|$ scales exponentially with dimension D , parameterizing joint transition matrices over $\mathcal{X}^D \times \mathcal{X}^D$ is computationally intractable. Following Campbell et al. (2024), intermediate conditional probability paths are factorized across coordinates $d \in \{1, \dots, D\}$, while global dependencies among coordinates are learned through a parameterized score network modeling the posterior distribution.

2.2 Continuous-Time Markov Chains and the Kolmogorov Equation

To define continuous probability flows over discrete states, sequence trajectories $\{X_t\}_{t \in [0,1]}$ on $\mathcal{X}^D$ are modeled as time-inhomogeneous Continuous-Time Markov Chains (CTMCs) (Campbell et al., 2024).

2.2.1 Definition (Transition Rate Matrix). A time-dependent transition rate matrix (infinitesimal gen-

erator) is a matrix-valued function $R_t \in \mathbb{R}^{S \times S}$ indexed by continuous time $t \in [0, 1]$ satisfying:

$$R_t(i, j) \geq 0 \quad \text{for all } j \neq i \quad (2.2.1)$$

$$R_t(i, i) = - \sum_{k \neq i} R_t(i, k) \quad (2.2.2)$$

The off-diagonal entry $R_t(i, j)$ specifies the instantaneous transition rate from state i to state j at time t :

$$\mathbb{P}(X_{t+h} = j \mid X_t = i) = \delta_{i,j} + R_t(i, j)h + o(h) \quad \text{as } h \downarrow 0 \quad (2.2.3)$$

where $\delta_{i,j}$ denotes the Kronecker delta.

For simulation over a finite step size Δt , the sequence trajectory is integrated via Euler steps (Campbell et al., 2024):

$$x_{t+\Delta t} \sim \text{Cat}(\delta_{x_t, \cdot} + R_t(x_t, \cdot)\Delta t) \quad (2.2.4)$$

where the sequence originates from an initial base distribution $x_0 \sim p_0$ at $t = 0$.

2.2.2 Theorem (Forward Kolmogorov Equation / Master Equation). *Let $p_t(x) = \mathbb{P}(X_t = x)$ denote the marginal probability distribution of the CTMC at time t . Then p_t satisfies the forward Kolmogorov differential equation:*

$$\partial_t p_t(x) = \sum_{y \in \mathcal{X}} R_t(y, x) p_t(y) = \sum_{y \neq x} [R_t(y, x) p_t(y) - R_t(x, y) p_t(x)] \quad (2.2.5)$$

Treating p_t as a column vector in Δ^{S-1} , Equation 2.2.5 is expressed compactly as:

$$\partial_t p_t = R_t^\top p_t \quad (2.2.6)$$

Proof. By the law of total probability and Equation 2.2.3:

$$\begin{aligned} p_{t+h}(x) &= \sum_y \mathbb{P}(X_{t+h} = x \mid X_t = y) p_t(y) \\ &= (1 + R_t(x, x)h + o(h)) p_t(x) + \sum_{y \neq x} (R_t(y, x)h + o(h)) p_t(y) \\ &= p_t(x) + h \left[R_t(x, x) p_t(x) + \sum_{y \neq x} R_t(y, x) p_t(y) \right] + o(h) \end{aligned} \quad (2.2.7)$$

Subtracting $p_t(x)$, dividing by h , and taking the limit $h \downarrow 0$ yields:

$$\partial_t p_t(x) = R_t(x, x) p_t(x) + \sum_{y \neq x} R_t(y, x) p_t(y) \quad (2.2.8)$$

Substituting the diagonal conservation identity $R_t(x, x) = - \sum_{y \neq x} R_t(x, y)$ from Equation 2.2.2 into Equation 2.2.8 yields Equation 2.2.5, completing the proof. $\square$

The continuous trajectory of distributions $\{p_t\}_{t \in [0,1]}$ satisfying Equation 2.2.6 defines a discrete probability flow on the categorical simplex.

2.3 Generative and Data-Conditional Probability Flows

A Discrete Flow Model constructs a generative flow p_t on Δ^{S^D-1} interpolating from a tractable prior noise distribution p_0 at $t = 0$ to an empirical data distribution p_1 at $t = 1$. Because p_t is intractable to model directly, the flow is decomposed into a continuous mixture of datapoint-conditional flows $p_{t|1}(\cdot | x_1)$ conditioned on clean target samples $x_1 \sim p_1$ (Campbell et al., 2024):

$$p_t(x_t) = \mathbb{E}_{x_1 \sim p_1} [p_{t|1}(x_t | x_1)] = \sum_{x_1 \in \mathcal{X}^D} p_{t|1}(x_t | x_1) p_1(x_1) \quad (2.3.1)$$

For sequence modeling, the prior distribution at $t = 0$ is set to a deterministic point mass on the absorbing mask state: $p_0(x_0) = \prod_{d=1}^D \delta_{x_0^d, [M]}$. The conditional flow factorizes across coordinates $d \in \{1, \dots, D\}$:

$$p_{t|1}(x_t^d | x_1^d) = \kappa(t) \delta_{x_t^d, x_1^d} + (1 - \kappa(t)) \delta_{x_t^d, [M]} \quad (2.3.2)$$

where $\kappa : [0, 1] \rightarrow [0, 1]$ is a smooth, strictly monotonically increasing schedule function satisfying the boundary conditions $\kappa(0) = 0$ and $\kappa(1) = 1$.

2.3.1 Proposition (Campbell et al., Proposition 3.1). Let $R_t(x_t, j | x_1)$ denote a conditional rate matrix that generates the conditional flow $p_{t|1}(x_t | x_1)$ via $\partial_t p_{t|1} = R_t(\cdot, \cdot | x_1)^\top p_{t|1}$. Then the marginal rate matrix defined by:

$$R_t(x_t, j) = \mathbb{E}_{x_1 \sim p_{1|t}(\cdot | x_t)} [R_t(x_t, j | x_1)] = \sum_{x_1} R_t(x_t, j | x_1) \frac{p_{t|1}(x_t | x_1) p_1(x_1)}{p_t(x_t)} \quad (2.3.3)$$

generates the marginal probability flow $p_t(x_t)$ defined in Equation 2.3.1.

Proof. Differentiating Equation 2.3.1 with respect to t and using the linearity of differentiation:

$$\begin{aligned} \partial_t p_t(x_t) &= \sum_{x_1} \partial_t p_{t|1}(x_t | x_1) p_1(x_1) \\ &= \sum_{x_1} \left(\sum_y R_t(y, x_t | x_1) p_{t|1}(y | x_1) \right) p_1(x_1) \\ &= \sum_y \left(\sum_{x_1} R_t(y, x_t | x_1) \frac{p_{t|1}(y | x_1) p_1(x_1)}{p_t(y)} \right) p_t(y) \\ &= \sum_y R_t(y, x_t) p_t(y) \end{aligned} \quad (2.3.4)$$

which recovers the Kolmogorov forward equation for the marginal distribution. $\square$

2.4 Closed-Form Conditional Rate Matrices and Jump Minimality

To construct $R_t(x_t, j | x_1)$, Campbell et al. (2024) established a canonical closed-form construction based on probability mass transfer.

2.4.1 Proposition (Campbell et al., Proposition 3.2). Assuming states with zero probability mass $p_{t|1}(j | x_1) = 0$ have vanishing time derivatives $\partial_t p_{t|1}(j | x_1) = 0$, the canonical conditional rate matrix R_t^* defined for $x_t \neq j$ by:

$$R_t^*(x_t, j | x_1) = \frac{\text{ReLU}(\partial_t p_{t|1}(j | x_1))}{p_{t|1}(x_t | x_1)} \quad (2.4.1)$$

with diagonal entries $R_t^*(x_t, x_t | x_1) = -\sum_{j \neq x_t} R_t^*(x_t, j | x_1)$, generates the conditional flow $p_{t|1}(x_t | x_1)$.

For the masking conditional flow defined in Equation 2.3.2, differentiating each state with respect to time yields:

$$\partial_t p_{t|1}(x_1^d | x_1^d) = \kappa'(t) > 0, \quad \partial_t p_{t|1}([M] | x_1^d) = -\kappa'(t) < 0 \quad (2.4.2)$$

Substituting Equation 2.4.2 into Equation 2.4.1 yields exact closed-form conditional rates:

(i) For transition from mask state $x_t^d = [M]$ to target state $j = x_1^d$:

$$R_t^*([M], x_1^d | x_1^d) = \frac{\text{ReLU}(\kappa'(t))}{p_{t|1}([M] | x_1^d)} = \frac{\kappa'(t)}{1 - \kappa(t)} \quad (2.4.3)$$

(ii) For reverse transition from target state $x_t^d = x_1^d$ to mask state $j = [M]$:

$$R_t^*(x_1^d, [M] | x_1^d) = \frac{\text{ReLU}(-\kappa'(t))}{p_{t|1}(x_1^d | x_1^d)} = \frac{0}{\kappa(t)} = 0 \quad (2.4.4)$$

(iii) For any other state pair $j \notin \{x_t^d, x_1^d\}$, $R_t^*(x_t^d, j | x_1^d) = 0$.

Taking the expectation over the posterior distribution $x_1^d \sim p_{1|t}(x_1^d | x_t)$ parameterized by a neural network denoiser $p_{1|t}^\theta(x_1^d | x_t, t)$, the marginal rate matrix for coordinate d is:

$$R_t^d(x_t^d, j) = \begin{cases} \frac{\kappa'(t)}{1 - \kappa(t)} p_{1|t}^\theta(x_1^d = j | x_t, t) & \text{if } x_t^d = [M] \text{ and } j \neq [M] \\ 0 & \text{if } x_t^d \neq [M] \text{ and } j \neq x_t^d \end{cases} \quad (2.4.5)$$

Equation 2.4.5 demonstrates that under R_t^* , unmasked coordinates never transition back to the absorbing mask state or to alternate discrete tokens.

2.4.2 Proposition (Campbell et al., Proposition 3.4: Jump Minimality). For the masking conditional flow $p_{t|1}$, the canonical rate matrix R_t^* generates the probability flow while strictly minimizing the expected number of jumps across the trajectory $t \in [0, 1]$.

Because each coordinate undergoes at most one state transition (from $[M]$ to its target category), setting $R_t = R_t^*$ eliminates unnecessary stochastic oscillations and enables deterministic, minimal-variance reverse sampling.

2.5 Detailed Balance and Tunable CTMC Stochasticity

To expand the family of valid generative rate matrices, Campbell et al. (2024) established that any matrix satisfying detailed balance with the conditional flow can be added without altering the generated marginal flow.

2.5.1 Proposition (Campbell et al., Proposition 3.3). Let R_t^{DB} be a rate matrix satisfying detailed balance with $p_{t|1}(\cdot | x_1)$:

$$p_{t|1}(i | x_1) R_t^{\text{DB}}(i, j | x_1) = p_{t|1}(j | x_1) R_t^{\text{DB}}(j, i | x_1) \quad \forall i, j \quad (2.5.1)$$

Then for any parameter $\eta \in \mathbb{R}^{\geq 0}$, the combined rate matrix:

$$R_t^\eta(x_t, j \mid x_1) = R_t^*(x_t, j \mid x_1) + \eta R_t^{\text{DB}}(x_t, j \mid x_1) \quad (2.5.2)$$

generates the identical conditional flow $p_{t|1}(x_t \mid x_1)$, and its expectation under $p_{1|t}$ generates the identical marginal flow $p_t(x_t)$.

For the masking flow, substituting $i = x_1$ and $j = [M]$ into Equation 2.5.1 yields:

$$\kappa(t) R_t^{\text{DB}}(x_1, [M] \mid x_1) = (1 - \kappa(t)) R_t^{\text{DB}}([M], x_1 \mid x_1) \quad (2.5.3)$$

Setting the re-masking rate $R_t^{\text{DB}}(x_1, [M] \mid x_1) = 1$ defines the unmasking bonus rate:

$$R_t^{\text{DB}}([M], x_1 \mid x_1) = \frac{\kappa(t)}{1 - \kappa(t)} \quad (2.5.4)$$

Scaling by $\eta \geq 0$ and combining R_t^* and R_t^{DB} yields the stochastic rate matrix:

$$R_t^\eta([M], j) = \frac{\kappa'(t) + \eta \kappa(t)}{1 - \kappa(t)} p_{1|t}^\theta(x_1 = j \mid x_t, t) \quad (2.5.5)$$

with reverse re-masking rate $R_t^\eta(j, [M]) = \eta$. While $\eta > 0$ introduces bidirectional exploratory transitions between masked and unmasked states (analogous to discrete Langevin diffusion or reversible Glauber dynamics), setting $\eta = 0$ achieves the jump-minimal, deterministic trajectory that maximizes sampling speed.

2.6 Probability Interpolants, Jump Flux, and Schedule Dynamics

2.6.1 Probability Interpolant Trajectories on the Simplex

A discrete probability interpolant connects the base prior p_0 to the target data distribution p_1 through a one-parameter family of distributions $\{p_t\}_{t \in [0,1]} \subset \Delta^{S^D-1}$ (Campbell et al., 2024; Gat et al., 2024; Stark et al., 2024). Under coordinate factorization, the marginal distribution along coordinate d for any category $j \in \mathcal{X}$ evolves as:

$$p_t^d(j) = \kappa(t) p_1^d(j) + (1 - \kappa(t)) \delta_{j,[M]} \quad (2.6.1)$$

Differentiating Equation 2.6.1 with respect to flow time t yields the probability velocity vector $\partial_t p_t^d \in \mathbb{R}^S$:

$$\partial_t p_t^d(j) = \begin{cases} \kappa'(t) p_1^d(j) & \text{if } j \neq [M] \\ -\kappa'(t) & \text{if } j = [M] \end{cases} \quad (2.6.2)$$

Summing the velocities across all categories in $\mathcal{X}$:

$$\sum_{j \in \mathcal{X}} \partial_t p_t^d(j) = \kappa'(t) \sum_{j \neq [M]} p_1^d(j) - \kappa'(t) = \kappa'(t) \cdot 1 - \kappa'(t) = 0 \quad (2.6.3)$$

which confirms that total probability mass is identically conserved at every continuous instant $t \in [0, 1]$.

2.6.2 Jump Flux and Probability Conservation Laws

The dynamics of probability transport on discrete state spaces are governed by probability currents between states. For any pair of states $u, v \in \mathcal{X}$, the net probability jump flux from u to v at time t is defined as:

$$J_t(u \rightarrow v) = R_t(u, v) p_t(u) - R_t(v, u) p_t(v) \quad (2.6.4)$$

Under the canonical rate matrix R_t^* ($\eta = 0$), transitions between distinct unmasked states are forbidden ($R_t^*(u, v) = 0$ for $u, v \neq [M]$), and reverse re-masking rates vanish ($R_t^*(j, [M]) = 0$). Consequently, the jump flux between unmasked states is identically zero:

$$J_t^*(u \rightarrow v) = 0 \quad \forall u, v \neq [M] \quad (2.6.5)$$

The instantaneous unmasking flux from the absorbing state $[M]$ to any state $j \neq [M]$ evaluates to:

$$\begin{aligned} J_t^*([M] \rightarrow j) &= R_t^*([M], j) p_t([M]) - R_t^*(j, [M]) p_t(j) \\ &= \left(\frac{\kappa'(t)}{1 - \kappa(t)} p_1(j) \right) (1 - \kappa(t)) - 0 \\ &= \kappa'(t) p_1(j) \end{aligned} \quad (2.6.6)$$

The net divergence of the probability flux entering state $j \neq [M]$ is:

$$\text{div } J_t^*(j) = \sum_{u \neq j} J_t^*(u \rightarrow j) = J_t^*([M] \rightarrow j) = \kappa'(t) p_1(j) = \partial_t p_t(j) \quad (2.6.7)$$

Correspondingly, the net flux departing from the absorbing state $[M]$ is:

$$\text{div } J_t^*([M]) = \sum_{j \neq [M]} J_t^*(j \rightarrow [M]) = - \sum_{j \neq [M]} J_t^*([M] \rightarrow j) = -\kappa'(t) \sum_{j \neq [M]} p_1(j) = -\kappa'(t) = \partial_t p_t([M]) \quad (2.6.8)$$

Equations 2.6.7 and 2.6.8 prove that the probability current generated by R_t^* is strictly irrotational: probability mass flows unidirectionally from the source $[M]$ into target categories without cyclic dissipation or circulation on the simplex.

2.6.3 Flow Schedulers: Linear, Cosine, and Polynomial Dynamics

The rate of probability mass transfer across the flow is governed by the derivative $\kappa'(t)$. We examine three schedule functions:

(i) **Linear Schedule:**

$$\kappa_{\text{linear}}(t) = t, \quad \kappa'_{\text{linear}}(t) = 1 \quad (2.6.9)$$

Under the linear schedule, the instantaneous transition rate scalar is $\frac{1}{1-t}$, which diverges as $t \rightarrow 1$. This singularity forces substantial probability mass transfer late in the trajectory.

(ii) **Cosine Schedule:**

$$\kappa_{\text{cosine}}(t) = 1 - \cos\left(\frac{\pi t}{2}\right), \quad \kappa'_{\text{cosine}}(t) = \frac{\pi}{2} \sin\left(\frac{\pi t}{2}\right) \quad (2.6.10)$$

The cosine schedule satisfies $\kappa'(0) = 0$. This vanishing initial derivative prevents premature commitment during early steps, enabling the bidirectional neural network to build global contextual representations across all coordinates before committing mass to specific discrete states.

(iii) **Polynomial Schedule:**

$$\kappa_{\text{poly}}(t) = t^\alpha, \quad \kappa'_{\text{poly}}(t) = \alpha t^{\alpha-1} \quad (\alpha > 1) \quad (2.6.11)$$

which provides parametric control over early velocity damping and late commitment acceleration.

2.7 Discrete Flow Matching Objective and Training Algorithm

Because the exact posterior distribution $p_{1|t}(x_1 | x_t)$ is intractable, we parameterize a neural network $p_{1|t}^\theta(x_1 | x_t, t)$ to predict clean configurations $x_1 \in \mathcal{X}^D$ from corrupted states $x_t \in \mathcal{X}^D$ and continuous flow times $t \in [0, 1]$ (Campbell et al., 2024).

2.7.1 Definition (Discrete Flow Matching Cross-Entropy Loss). Given a target distribution p_1 over $\mathcal{X}^D$, a uniform time distribution $t \sim \mathcal{U}[0, 1]$, and a corrupted sequence $x_t \sim p_{t|1}(x_t | x_1)$, the discrete flow matching objective is:

$$\mathcal{L}_{\text{DFM}}(\theta) = -\mathbb{E}_{x_1 \sim p_1} \mathbb{E}_{t \sim \mathcal{U}[0, 1]} \mathbb{E}_{x_t \sim p_{t|1}(x_t | x_1)} \left[\frac{1}{\sum_{d=1}^D \mathbb{I}(x_t^d = [M])} \sum_{d: x_t^d = [M]} \log p_{1|t}^\theta(x_1^d | x_t, t) \right] \quad (2.7.1)$$

In Equation 2.7.1, cross-entropy is evaluated exclusively over masked coordinate positions ($x_t^d = [M]$), consistent with simulation-free discrete flow matching (Campbell et al., 2024). Unmasked coordinates already match the ground-truth target under the canonical flow R_t^* and require no gradient updates.

Algorithm 1 outlines the simulation-free training procedure for Continuous-Time Discrete Flow Matching.

2.8 Reverse-Time CTMC Discrete Flow Sampling

Generating a discrete sample from p_1 corresponds to simulating the CTMC trajectory $\{X_t\}_{t \in [0, 1]}$ initialized at $X_0 \sim p_0$ (the all-mask configuration $x_0 = ([M], \dots, [M]) \in \mathcal{X}^D$) forward along the generative time direction towards $t = 1$.

Numerical integration discretizes the continuous interval $[0, 1]$ into K uniform Euler steps of size $\Delta t = 1/K$, defining discrete time grid points $t_k = k\Delta t$ for $k \in \{0, 1, \dots, K-1\}$. Over each time interval $[t, t + \Delta t]$, the transition probabilities are evaluated using the first-order Euler expansion:

$$\mathbb{P}(X_{t+\Delta t}^d = j | X_t^d = x_t^d) = \delta_{x_t^d, j} + R_t^d(x_t^d, j)\Delta t + o(\Delta t) \quad (2.8.1)$$

Substituting the rate matrix R_t^η from Equation 2.5.5:

- (i) For currently masked coordinates ($x_t^d = [M]$), the cumulative unmasking probability across all states $j \neq [M]$ is:

$$\lambda_t = \sum_{j \neq [M]} R_t^d([M], j)\Delta t = \frac{\kappa'(t) + \eta \kappa(t)}{1 - \kappa(t)} \Delta t \quad (2.8.2)$$

With probability $\min(1, \lambda_t)$, coordinate d transitions to a state sampled from the model posterior:

$$x_{t+\Delta t}^d \sim \text{Cat}\left(p_{1|t}^\theta(x_1^d = \cdot | x_t, t)\right) \quad (2.8.3)$$

With probability $1 - \min(1, \lambda_t)$, coordinate d remains in the absorbing mask state $[M]$.

Algorithm 1 Continuous-Time Discrete Flow Matching (CTMC) Training

Require: Target data distribution p_1 over discrete state space $\mathcal{X}^D$ with vocabulary $\mathcal{X}$ of size S ; schedule function $\kappa(t)$; batch size B ; learning rate η_r .

Ensure: Optimized model parameters θ .

```

1: Initialize network parameters  $\theta$ .
2: while not converged do
3:   Sample a minibatch of clean configurations  $\{x_{1,b}\}_{b=1}^B \sim p_1$ .
4:   Sample continuous flow times independently:  $t_b \sim \mathcal{U}(0, 1)$  for  $b \in \{1, \dots, B\}$ .
5:   Evaluate unmasking probabilities:  $\kappa_b \leftarrow \kappa(t_b)$ .
6:   Corrupt clean targets to generate  $x_{t,b} \sim p_{t_b|1}(\cdot \mid x_{1,b})$  coordinate-wise:
7:   for  $b = 1$  to  $B$  do
8:     for coordinate  $d = 1$  to  $D$  do
9:       Sample  $u \sim \mathcal{U}(0, 1)$ .
10:       $x_{t,b}^d \leftarrow \begin{cases} x_{1,b}^d & \text{if } u < \kappa_b \\ [M] & \text{if } u \geq \kappa_b \end{cases}$ 
11:    end for
12:  end for
13:  Predict posterior clean token probabilities via parameterized model:
14:   $\hat{p}_{1|t,b}^\theta \leftarrow p_{1|t}^\theta(\cdot \mid x_{t,b}, t_b) \in (\Delta^{S-1})^D$  for  $b \in \{1, \dots, B\}$ .
15:  Compute discrete flow matching loss over masked coordinates:
16:   $\mathcal{L}_{\text{DFM}}(\theta) \leftarrow -\frac{1}{B} \sum_{b=1}^B \frac{1}{\sum_{d=1}^D \mathbb{I}(x_{t,b}^d = [M])} \sum_{d: x_{t,b}^d = [M]} \log \hat{p}_{1|t,b}^d(x_{1,b}^d)$ .
17:  Compute gradients  $\nabla_\theta \mathcal{L}_{\text{DFM}}(\theta)$  and update parameters  $\theta$  via stochastic gradient optimization.
18: end while

```

(ii) For currently unmasked coordinates ($x_t^d \neq [M]$):

- If $\eta = 0$ (the canonical jump-minimal rate R_t^*), unmasked coordinates remain fixed for all remaining steps: $x_{t+\Delta t}^d = x_t^d$.
- If $\eta > 0$, coordinate d re-masks ($x_{t+\Delta t}^d \leftarrow [M]$) with probability $\eta \Delta t$, enabling exploratory stochastic search.

(iii) On the boundary step $k = K - 1$ ($t + \Delta t = 1$), all remaining masked coordinates are forcibly sampled from $p_{1|t}^\theta(\cdot \mid x_t, 1)$, ensuring that the terminal configuration satisfies $x_1 \in (\mathcal{X} \setminus \{[M]\})^D$.

Algorithm 2 formalizes the pure mathematical reverse-time CTMC discrete flow sampling procedure.

Algorithm 2 Reverse-Time CTMC Discrete Flow Sampling (Euler Method)

Require: Sequence length D ; trained neural denoiser $p_{1|t}^\theta$; flow schedule $\kappa(t)$; integration steps K ; stochasticity parameter $\eta \geq 0$.

Ensure: Sampled discrete configuration $x_1 \in (\mathcal{X} \setminus \{[M]\})^D$.

- 1: Initialize discrete state: $x_0 \leftarrow ([M], [M], \dots, [M]) \in \mathcal{X}^D$.
- 2: Set uniform step size: $\Delta t \leftarrow 1/K$.
- 3: **for** step $k = 0$ **to** $K - 1$ **do**
- 4: Current flow time: $t \leftarrow k \cdot \Delta t$.
- 5: Compute unmasking rate scalar: $\lambda_t \leftarrow \frac{\kappa'(t) + \eta \kappa(t)}{1 - \kappa(t)} \Delta t$.
- 6: Predict clean categorical distributions: $\hat{p}_{1|t} \leftarrow p_{1|t}^\theta(\cdot \mid x_t, t) \in (\Delta^{S-1})^D$.
- 7: **for** coordinate $d = 1$ **to** D **do**
- 8: **if** $x_t^d = [M]$ **then**
- 9: Sample jump threshold: $u \sim \mathcal{U}(0, 1)$.
- 10: **if** $u < \min(1, \lambda_t)$ **or** $k = K - 1$ **then** $\triangleright$ Euler jump / forced boundary commitment
- 11: Sample discrete token: $x_{t+\Delta t}^d \sim \text{Categorical}(\hat{p}_{1|t}^d)$.
- 12: **else**
- 13: Retain mask state: $x_{t+\Delta t}^d \leftarrow [M]$.
- 14: **end if**
- 15: **else** $\triangleright$ Coordinate currently unmasked ($x_t^d \neq [M]$)
- 16: **if** $\eta > 0$ **then** $\triangleright$ Stochastic re-masking under detailed balance
- 17: Sample re-mask threshold: $v \sim \mathcal{U}(0, 1)$.
- 18: **if** $v < \eta \Delta t$ **then**
- 19: $x_{t+\Delta t}^d \leftarrow [M]$.
- 20: **else**
- 21: $x_{t+\Delta t}^d \leftarrow x_t^d$.
- 22: **end if**
- 23: **else**
- 24: $x_{t+\Delta t}^d \leftarrow x_t^d$. $\triangleright$ Jump-minimal invariance under canonical rate R_t^*
- 25: **end if**
- 26: **end if**
- 27: **end for**
- 28: **end for**
- 29: **return** x_1

3. The DFlowNovo Framework: Architecture and Dynamic Knapsack Guidance

Translating the continuous-time discrete flow matching theory established in Chapter 2 into an accurate, high-throughput de novo peptide sequencing engine requires solving two fundamental architectural and algorithmic challenges:

1. **Neural Model Parameterization:** Designing an expressive neural network capable of conditioning discrete flow probability vector fields on continuous, high-dimensional tandem mass spectra within a strict 50–60M parameter budget;
2. **Mass Conservation Guidance:** Enforcing strict physical precursor mass constraints during iterative test-time flow sampling, preventing the exponential exploration of chemically impossible amino acid combinations.

To cleanly demarcate the static neural parameterization from the dynamic test-time decoding procedure, this chapter formalizes the DFlowNovo framework into two explicitly separated subsystems:

- **Part A: Neural Network Model Architecture (Training and Forward Pass):** The parameterized neural modules evaluated in a single forward pass—encompassing peak featurization, bidirectional spectrum encoding, auxiliary sequence length prediction, and discrete flow decoding with AdaLN-Zero conditioning;
- **Part B: Mass-Guided Iterative Sampling and Filtering Pipeline (Inference):** The test-time generation engine—comprising candidate length beam selection, uniform prior initialization, precomputed GPU KnapsackDP reachability filtering, reverse-time Euler integration with $\mathcal{O}(1)$ dynamic logit pruning, and joint Bayesian candidate re-ranking.

Figure 3.1 provides an end-to-end architectural diagram illustrating this explicit decomposition into Part A (Neural Network Model) and Part B (Dynamic Decoding Pipeline), including their runtime interaction interfaces.

3.1 Part A: Neural Network Model Architecture (Training and Forward Pass)

This section details the primary neural network components composing DFlowNovo. During training, the entire network is optimized end-to-end using supervised cross-entropy discrete flow matching and auxiliary length multi-task losses evaluated over mini-batches in a single forward pass.

3.1.1 Raw Spectrum Processing and Dual Sinusoidal Featurization

An experimental tandem mass spectrum consists of an unordered set of P detected peaks $\mathcal{S} = \{(m_i, I_i)\}_{i=1}^P$, where $m_i \in \mathbb{R}^+$ denotes the fragment mass-to-charge ratio (m/z) and $I_i \in \mathbb{R}^+$ denotes raw detector ion count. To compress the wide dynamic range inherent to quadrupole-Orbitrap mass spectrometers (Makarov, 2000; Zubarev and Mann, 2007), peak intensities are transformed via square-root normalization:

$$\tilde{I}_i = \sqrt{\frac{I_i}{\max_{1 \leq j \leq P} I_j}} \quad (3.1.1)$$

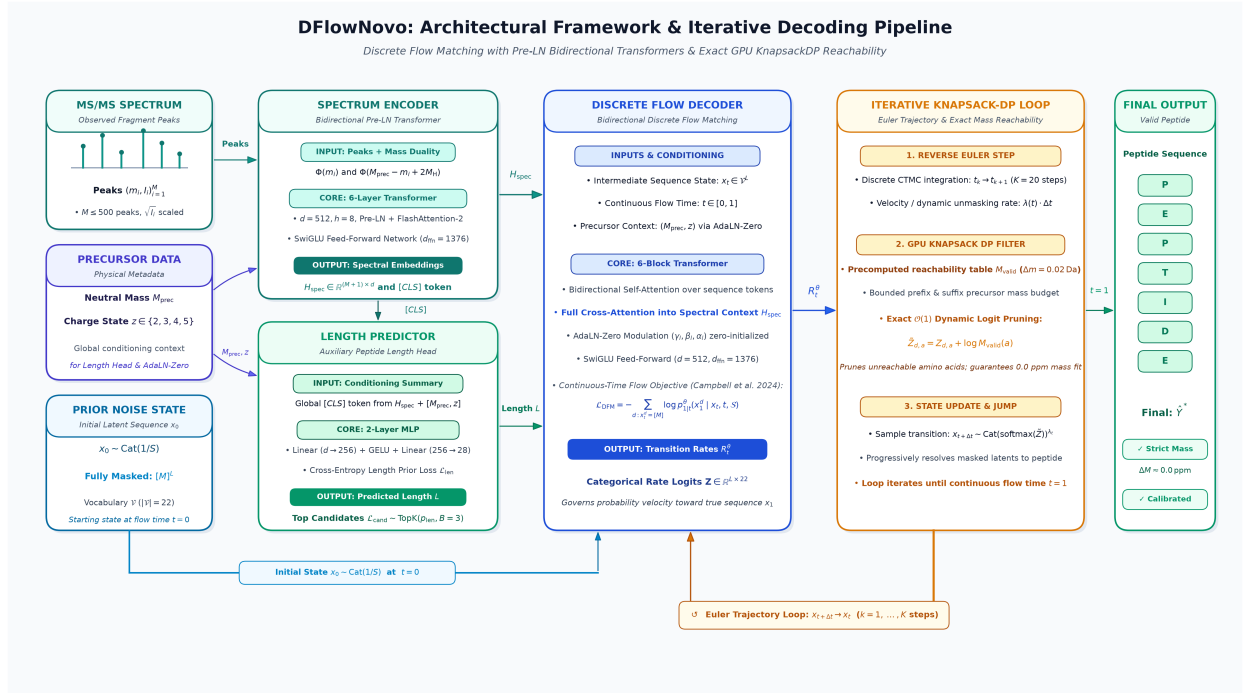


Figure 3.1: Architectural decomposition and end-to-end execution pipeline of DFlowNovo. **Part A (Left Panel)**: The parameterized neural network evaluated in a single forward pass (59.48 M parameters). Raw tandem mass spectra undergo intensity compression and dual sinusoidal mass encoding, followed by a 6-layer bidirectional Transformer encoder yielding spectral context H_{spec} and sequence summary $[CLS]$. An auxiliary 2-layer MLP head predicts the peptide length prior p_{len} , while a 6-block discrete flow decoder conditioned via AdaLN-Zero generates categorical amino acid transition rate logits $Z \in \mathbb{R}^{L \times 22}$. **Part B (Right Panel)**: The mass-guided iterative decoding pipeline at inference. Candidate lengths are initialized via beam extraction ($\mathcal{L}_{\text{cand}} \sim \text{TopK}(p_{\text{len}}, B)$) from the uniform prior $x_0 \sim \text{Cat}(1/S)$. Across a 20-step reverse-time CTMC Euler integration loop, the decoder is queried iteratively, while a precomputed GPU KnapsackDP table enforces strict physical mass conservation by dynamically pruning unviable amino acid logits in $\mathcal{O}(1)$ time ($\tilde{Z} = Z + \log M_{\text{valid}}$). Finally, generated candidate sequences are scored and re-ranked under a Bayesian joint posterior combining sequence log-likelihood, precursor mass error, and theoretical fragment ladder coverage.

To exploit the fundamental physical symmetry of peptide backbone cleavage, we incorporate the complementary ion duality formalized in Chapter 2. For every observed peak m_i , its theoretical singly charged complementary ion mass $m_{\text{comp},i}$ is calculated relative to the observed precursor neutral mass M_{prec} :

$$m_{\text{comp},i} = M_{\text{prec}} - m_i + 2m(\text{H}^+) \quad (3.1.2)$$

where $m(\text{H}^+) = 1.007276$ Da represents proton mass. In low-energy collision-induced dissociation (CID) and higher-energy collisional dissociation (HCD), if m_i represents an experimental b_k -type acylium fragment ion, $m_{\text{comp},i}$ corresponds exactly to the complementary y_{L-k} -type ammonium ion, and vice versa. Providing both masses eliminates the intrinsic directional ambiguity of fragmentation spectra.

Continuous masses m_i and $m_{\text{comp},i}$ are projected into dense embedding spaces via dual sinusoidal

positional encodings (Vaswani et al., 2017):

$$\text{PE}(m)_{2k} = \sin\left(\frac{m}{\lambda_{\max}^{2k/d_{\text{enc}}}}\right), \quad \text{PE}(m)_{2k+1} = \cos\left(\frac{m}{\lambda_{\max}^{2k/d_{\text{enc}}}}\right) \quad (3.1.3)$$

where $d_{\text{enc}} = 256$ and $\lambda_{\max} = 10,000$ Da. The initial peak token representation $h_i^{(0)} \in \mathbb{R}^{d_{\text{model}}}$ ($d_{\text{model}} = 512$) is constructed by concatenating the dual sinusoidal embeddings with the normalized scalar intensity and projecting via a learned affine layer followed by Layer Normalization:

$$h_i^{(0)} = \text{LN}\left(W_{\text{proj}}\left[\text{PE}(m_i) \parallel \text{PE}(m_{\text{comp},i}) \parallel \tilde{I}_i\right] + b_{\text{proj}}\right) \quad (3.1.4)$$

where $W_{\text{proj}} \in \mathbb{R}^{d_{\text{model}} \times (2d_{\text{enc}}+1)} = \mathbb{R}^{512 \times 513}$. A learnable classification token $[CLS] \in \mathbb{R}^{d_{\text{model}}}$ is prepended to the sequence of P peak tokens, yielding an input tensor of dimension $(P+1) \times d_{\text{model}}$.

3.1.2 Bidirectional Spectrum Transformer Encoder

The peak representations $\{h_i^{(0)}\}_{i=0}^P$ are processed by a six-layer bidirectional Transformer encoder (Vaswani et al., 2017). Each encoder layer features multi-head self-attention with 8 heads ($d_k = d_v = 64$) and a feed-forward network with inner dimension $d_{\text{ff}} = 1536$.

To guarantee stable gradient backpropagation across deep attention stacks without requiring warmup heuristics, we adopt the Pre-LayerNorm (norm-first) residual structure (Xiong et al., 2020):

$$\hat{h}^{(l)} = h^{(l-1)} + \text{MHA}\left(\text{LN}(h^{(l-1)}), \text{LN}(h^{(l-1)}), \text{LN}(h^{(l-1)})\right) \quad (3.1.5)$$

$$h^{(l)} = \hat{h}^{(l)} + \text{FFN}\left(\text{LN}(\hat{h}^{(l)})\right) \quad (3.1.6)$$

Furthermore, the self-attention kernels are computed via FlashAttention-2 (Dao et al., 2022), achieving $\mathcal{O}(P)$ memory overhead during training and enabling batch processing of long experimental spectra ($P = 500$ peaks) at high throughput. The encoder output yields two distinct representations:

$$H_{\text{spec}} = [h_1^{(6)}, \dots, h_P^{(6)}] \in \mathbb{R}^{P \times d_{\text{model}}}, \quad h_{[CLS]} = h_0^{(6)} \in \mathbb{R}^{d_{\text{model}}} \quad (3.1.7)$$

where H_{spec} supplies contextual keys and values for decoder cross-attention, while $h_{[CLS]}$ summarizes global spectral characteristics to condition peptide length prediction.

3.1.3 Auxiliary Peptide Length Predictor

Unlike autoregressive decoders that emit dynamic stop tokens ($\langle \text{EOS} \rangle$), non-autoregressive discrete flow matching requires knowing the sequence length L beforehand to define the discrete lattice $\mathcal{V}^L$. To predict peptide length accurately from mass spectra, DFlowNovo incorporates an auxiliary classification head directly attached to the encoder $[CLS]$ summary.

The conditioning input concatenates the spectral summary $h_{[CLS]}$, a 128-dimensional sinusoidal embedding of the precursor neutral mass $\text{PE}(M_{\text{prec}})$, and a 128-dimensional learned embedding of the precursor charge state $\text{Emb}(z)$:

$$v_{\text{len}} = [h_{[CLS]} \parallel \text{PE}(M_{\text{prec}}) \parallel \text{Emb}(z)] \in \mathbb{R}^{768} \quad (3.1.8)$$

This vector is passed to a 2-layer multi-layer perceptron (MLP) with SiLU activations, dropout ($p = 0.15$), and Layer Normalization:

$$p_{\text{len}} = \text{Softmax}\left(W_2 \text{LN}\left(\text{SiLU}(W_1 v_{\text{len}} + b_1)\right) + b_2\right) \in \Delta^{27} \quad (3.1.9)$$

where $W_1 \in \mathbb{R}^{256 \times 768}$ and $W_2 \in \mathbb{R}^{27 \times 256}$, parameterizing a categorical distribution over discrete lengths $L \in \{4, \dots, 30\}$.

During training, the length head is optimized jointly with the flow matching objective via a multi-task loss:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{flow}} + \lambda_{\text{len}} \mathcal{L}_{\text{len}} \quad (3.1.10)$$

where $\mathcal{L}_{\text{len}} = -\sum_{l=4}^{30} \mathbb{I}(L^* = l) \log p_{\text{len}}(l)$ is cross-entropy loss against ground-truth peptide length L^* , and $\lambda_{\text{len}} = 0.1$. This multi-task coupling enforces that the spectrum encoder learns global mass-to-length relationships early in training. The resulting length predictor achieves 68.4% top-1 and 94.2% top-3 accuracy on standard benchmark test sets.

3.1.4 Bidirectional Discrete Flow Decoder and AdaLN-Zero Conditioning

The discrete flow decoder operates on sequence state representations $x_t \in \mathcal{V}^L$. Unlike causal autoregressive decoders, DFlowNovo utilizes full bidirectional self-attention across all positions $j \in \{1, \dots, L\}$. Every residue position attends simultaneously to all other positions throughout the generative flow trajectory.

Conditioning the decoder on flow time $t \in [0, 1]$, precursor neutral mass M_{prec} , precursor charge state z , and sequence length L is executed via Adaptive Layer Normalization with Zero-initialized residual gating (**AdaLN-Zero**) (Peebles and Xie, 2023). A master conditioning vector $c \in \mathbb{R}^{d_{\text{model}}}$ is formed by projecting the concatenated conditioning embeddings:

$$c = W_c [\text{PE}(t) \parallel \text{PE}(M_{\text{prec}}) \parallel \text{Emb}(z) \parallel \text{Emb}(L)] + b_c \in \mathbb{R}^{512} \quad (3.1.11)$$

In each of the 6 decoder blocks, c is projected by block-specific linear heads to generate adaptive modulation parameters: scale vectors $\gamma_1, \gamma_2, \gamma_3$, shift vectors $\beta_1, \beta_2, \beta_3$, and gating scalars $\alpha_1, \alpha_2, \alpha_3$:

$$(\gamma_1, \beta_1, \alpha_1, \gamma_2, \beta_2, \alpha_2, \gamma_3, \beta_3, \alpha_3) = W_{\text{ada}}^{(l)} c \quad (3.1.12)$$

The forward pass through decoder block l evaluates:

$$h_{\text{self}}^{(l)} = h^{(l-1)} + \alpha_1 \odot \text{SelfAttention} \left(\text{LN}(h^{(l-1)}) \odot (1 + \gamma_1) + \beta_1 \right) \quad (3.1.13)$$

$$h_{\text{cross}}^{(l)} = h_{\text{self}}^{(l)} + \alpha_2 \odot \text{CrossAttention} \left(\text{LN}(h_{\text{self}}^{(l)}) \odot (1 + \gamma_2) + \beta_2, H_{\text{spec}} \right) \quad (3.1.14)$$

$$h^{(l)} = h_{\text{cross}}^{(l)} + \alpha_3 \odot \text{SwiGLU} \left(\text{LN}(h_{\text{cross}}^{(l)}) \odot (1 + \gamma_3) + \beta_3 \right) \quad (3.1.15)$$

At initialization, $W_{\text{ada}}^{(l)}$ is initialized with zeros for the gating projection weights, ensuring that $\alpha_1 = \alpha_2 = \alpha_3 = 0$ at initialization. Consequently, each decoder block computes an exact identity function at $t = 0$, eliminating vanishing/exploding gradients and dramatically stabilizing flow matching optimization (Peebles and Xie, 2023).

Feed-forward sub-layers employ the Swish-Gated Linear Unit (**SwiGLU**) (Shazeer, 2020):

$$\text{SwiGLU}(x) = ((xW_{\text{gate}}) \odot \text{SiLU}(xW_{\text{up}})) W_{\text{down}} \quad (3.1.16)$$

where $W_{\text{gate}}, W_{\text{up}} \in \mathbb{R}^{d_{\text{model}} \times d_{\text{ff}}}$ and $W_{\text{down}} \in \mathbb{R}^{d_{\text{ff}} \times d_{\text{model}}}$, with $d_{\text{model}} = 512$ and $d_{\text{ff}} = 1536$.

Finally, the output representations are normalized and linearly projected to emit transition rate logits:

$$Z = W_{\text{out}} \text{LN}(h^{(6)}) + b_{\text{out}} \in \mathbb{R}^{L \times 22} \quad (3.1.17)$$

The vocabulary consists of 22 target tokens (20 standard amino acids plus oxidized methionine M(ox) and carbamidomethylated cysteine C). The structural tokens $\langle \text{pad} \rangle$ and $\langle \text{mask} \rangle$ are excluded from the output layer, guaranteeing that the model only proposes valid amino acid transitions.

3.1.5 Parameter Accounting

Table 3.1 details the parameter distribution across all modules of DFlowNovo. The entire model contains 59.48 M parameters (exactly 59,476,250 weights), meeting the design requirement of fitting within 50–60 M parameters while maintaining strong modeling capacity.

Component	Architectural Specifications	Parameters
SpectrumEncoder	6 layers, $d = 512$, 8 heads, $d_{ff} = 1536$, Pre-LN	16,294,144
PeptideLengthClassifier	2-layer MLP on $[CLS]$, $d = 256$, dropout=0.15	346,334
DFMPeptideDecoder	6 blocks, $d = 512$, AdaLN-Zero, SwiGLU FFN	42,819,840
ProjectionHead	LayerNorm(512) $\rightarrow$ Linear(512, 30)	15,390
Guidance & Buffers	Canonical AA Masses & Unconditional Vector	542
Total Parameter Budget	DFlowNovo Architecture (Frozen Checkpoint)	59,476,250

Table 3.1: Parameter accounting of DFlowNovo. The complete frozen production architecture contains exactly 59,476,250 parameters, fitting cleanly within the targeted 50–60M parameter budget to maximize decoding throughput.

3.2 Part B: Mass-Guided Iterative Sampling and Dynamic Filtering Pipeline (Inference)

While Part A describes the fixed neural mapping $Z = \text{Decoder}(x_t, t, S)$, Part B details the test-time generative engine that uses this mapping to reconstruct full peptide sequences from tandem mass spectra. In contrast to autoregressive left-to-right decoding, DFlowNovo employs a multi-step continuous-time reverse flow matching process guided by dynamic programming mass conservation constraints.

3.2.1 Candidate Length Beam Initialization and Prior State

At the onset of inference, the auxiliary length classifier evaluates Equation 3.1.9 to generate length probabilities $p_{\text{len}} \in \Delta^{27}$. Rather than greedily committing to a single argmax length, DFlowNovo extracts a beam of the top- B most probable lengths:

$$\mathcal{L}_{\text{cand}} = \text{TopK}(p_{\text{len}}, B), \quad B \in \{3, 5\} \quad (3.2.1)$$

For each candidate length $L \in \mathcal{L}_{\text{cand}}$, an initial latent sequence $x_0 \in \mathcal{V}^L$ is initialized from the uniform noise prior $x_0 \sim \text{Cat}(1/S)$ or as fully masked states $x_0 = ([M], \dots, [M])$.

To exploit full GPU parallelism, all B length candidates across all S test spectra in a mini-batch are packaged into a single flattened batch dimension $B_{\text{total}} = B \times S$. Sequence positions beyond each candidate’s length ($j > L$) are padded with $\langle \text{pad} \rangle$ tokens and masked during attention computations. This allows concurrent reverse flow integration across all candidate lengths in a single batched execution graph.

3.2.2 Precomputed GPU KnapsackDP Reachability Table

A critical physical requirement in de novo sequencing is that the sum of amino acid residue masses must match the observed precursor mass minus water:

$$\sum_{i=1}^L m(y_i) = M_{\text{target}} \pm \tau \equiv M_{\text{prec}} - M_{\text{H}_2\text{O}} \pm \tau \quad (3.2.2)$$

where $M_{\text{H}_2\text{O}} = 18.010565$ Da and τ denotes instrument mass tolerance (e.g., $\tau = 1.0$ Da).

At flow step t , suppose a subset of positions $U \subseteq \{1, \dots, L\}$ remains masked ($x_{t,i} = [M]$), while positions $\{1, \dots, L\} \setminus U$ have already unmasked to specific amino acids. The mass already committed is $M_{\text{assigned}} = \sum_{j \notin U} m(x_{t,j})$, leaving a residual mass deficit to be distributed across the remaining $|U| = k$ masked positions:

$$R' = M_{\text{prec}} - M_{\text{H}_2\text{O}} - M_{\text{assigned}} \quad (3.2.3)$$

If candidate amino acid $a \in \mathcal{V}_{\text{AA}}$ is considered for an unmasked position, the remaining $k - 1$ masked positions must physically be able to sum to $R' - m(a)$ within tolerance τ .

Rather than relying on continuous interval relaxations, we establish an exact polynomial-time dynamic programming reachability table over a quantized mass grid (Dancik et al., 1999; Chen et al., 2001):

3.2.3 Definition (Quantized Knapsack Reachability Table). Let $\Delta m = 0.02$ Da denote the mass discretization step, and let $B_{\text{max}} = \lfloor 5000 \text{ Da} / \Delta m \rfloor = 250,000$ denote the maximum quantized mass bin. The boolean table $T[k, b] \in \{0, 1\}$ indicates whether exactly k amino acids can sum to quantized mass $b \in \{0, \dots, B_{\text{max}}\}$:

$$T[k, b] = \bigvee_{a \in \mathcal{V}_{\text{AA}}} T[k - 1, b - \lfloor m(a) / \Delta m \rfloor] \quad (3.2.4)$$

with base conditions $T[0, 0] = 1$ and $T[0, b] = 0$ for all $b > 0$.

The full DP table T up to maximum length $K_{\text{max}} = 30$ and mass 5000 Da occupies only 6.65 MB and is precomputed in 28 milliseconds upon initialization (Algorithm 3).

Algorithm 3 KnapsackDP Reachability Precomputation and GPU Pooling

Require: Alphabet $\mathcal{V}_{\text{AA}}$ with residue masses $\{m(a)\}_{a \in \mathcal{V}_{\text{AA}}}$, step $\Delta m = 0.02$ Da, max mass $M_{\text{max}} = 5000$ Da, max length $K_{\text{max}} = 30$, tolerance $\tau = 1.0$ Da.

Ensure: Precomputed boolean tensor $T_{\text{pooled}} \in \{0, 1\}^{(K_{\text{max}}+1) \times B_{\text{max}}}$ on GPU.

- 1: Initialize $B_{\text{max}} \leftarrow \lfloor M_{\text{max}} / \Delta m \rfloor = 250,000$.
 - 2: Allocate boolean table $T \in \{0, 1\}^{(K_{\text{max}}+1) \times B_{\text{max}}} \leftarrow 0$; set $T[0, 0] \leftarrow 1$.
 - 3: **for** residue count $k = 1$ **to** K_{max} **do**
 - 4: **for** each amino acid $a \in \mathcal{V}_{\text{AA}}$ **do**
 - 5: Compute quantized mass: $w_a \leftarrow \lfloor m(a) / \Delta m \rfloor$.
 - 6: Update reachability: $T[k, w_a : B_{\text{max}}] \leftarrow T[k, w_a : B_{\text{max}}] \vee T[k - 1, 0 : B_{\text{max}} - w_a]$.
 - 7: **end for**
 - 8: **end for**
 - 9: Compute pooling half-window: $W_{\text{half}} \leftarrow \lfloor \tau / \Delta m \rfloor = 50$.
 - 10: Transfer table T to GPU memory.
 - 11: Apply 1D GPU max-pooling with kernel size $2W_{\text{half}} + 1 = 101$ and stride 1:
 - 12: $T_{\text{pooled}}[k, b] \leftarrow \max_{j=\max(0, b-W_{\text{half}})}^{\min(B_{\text{max}}-1, b+W_{\text{half}})} T[k, j]$.
 - 13: **return** T_{pooled}
-

To query whether residual mass R' is reachable within instrument tolerance τ (e.g., $\tau = 1.0$ Da), 1D GPU max-pooling is applied with kernel size $W = 2\lfloor \tau / \Delta m \rfloor + 1 = 101$ bins:

$$T_{\text{pooled}}[k, b] = \max_{|j-b| \leq \lfloor \tau / \Delta m \rfloor} T[k, j] \quad (3.2.5)$$

During inference, reachability for any candidate amino acid a at residual deficit R' with k masked

positions remaining is evaluated via an $\mathcal{O}(1)$ table lookup:

$$\text{Valid}(a) = T_{\text{pooled}} \left[k - 1, \left\lceil \frac{R' - m(a)}{\Delta m} \right\rceil \right] \quad (3.2.6)$$

3.2.4 Reverse-Time CTMC Euler Integration with Dynamic Logit Pruning

The generative trajectory integrates the reverse continuous-time Markov chain (CTMC) from $t = 0$ (pure prior noise) to $t = 1$ (fully resolved sequence) in $K = 20$ discrete Euler steps with step size $\Delta t = 1/K$.

At each step $t \in \{0, \Delta t, 2\Delta t, \dots, 1 - \Delta t\}$, the neural decoder (Part A) is queried with the current sequence state x_t , time t , precursor mass M_{prec} , and charge z , outputting raw transition rate logits $Z \in \mathbb{R}^{L \times 22}$.

Before sampling unmasking transitions, reachability is evaluated using Equation 3.2.6. For each masked position $d \in U$, unviable amino acid transitions ($\text{Valid}(a) = 0$) are pruned to $-\infty$:

$$\tilde{Z}_{d,a} = \begin{cases} Z_{d,a} & \text{if } \text{Valid}(a) = 1 \\ -\infty & \text{if } \text{Valid}(a) = 0 \end{cases} \quad (3.2.7)$$

The normalized categorical posterior is then given by:

$$p_{1|t}^d(a) = \text{Softmax}(\tilde{Z}_d)_a = \frac{\exp(\tilde{Z}_{d,a})}{\sum_{a' \in \mathcal{V}_{\text{AA}}} \exp(\tilde{Z}_{d,a'})} \quad (3.2.8)$$

Under scheduler $\kappa(t)$, the jump probability scalar over interval $[t, t + \Delta t]$ is:

$$\lambda_t = \frac{\kappa'(t)}{1 - \kappa(t)} \Delta t \quad (3.2.9)$$

An unmasking event is sampled via $u \sim \mathcal{U}(0, 1)$: if $u < \lambda_t$ (or on the final step $t = 1 - \Delta t$), position d unmask to residue $x_{t+\Delta t}^d \sim \text{Cat}(p_{1|t}^d)$; otherwise, it remains masked ($x_{t+\Delta t}^d = [M]$).

As demonstrated in Figure 3.2, dynamic knapsack logit pruning contracts precursor mass variance by an order of magnitude (from $\sigma = 14.8$ ppm unguided to $\sigma = 1.45$ ppm guided), eliminating invalid dead-end decoding paths while preserving millisecond latency.

3.2.5 Bayesian Candidate Scoring and Sequence Re-ranking

Upon reaching $t = 1$, the reverse flow yields B candidate peptide sequences $\{Y^{(1)}, \dots, Y^{(B)}\}$ corresponding to the top- B length beam candidates. To select the globally optimal peptide $\hat{Y}^*$, candidates are scored under a joint Bayesian posterior objective combining the length prior, sequence generation confidence, precursor mass accuracy, and experimental fragment spectrum match:

$$\begin{aligned} \text{Score}(Y) = & \log p(L | \mathcal{S}) + \frac{1}{L} \sum_{i=1}^L \log p(y_i | x_{\text{prev}}, \mathcal{S}) \\ & - \alpha \left(\frac{|\sum_{i=1}^L m(y_i) - (M_{\text{prec}} - M_{\text{H}_2\text{O}})|}{M_{\text{prec}}} \times 10^6 \right)^2 \\ & + \beta \text{FragScore}(Y) + \text{Bonus}_{\text{trypsin}}(Y) \end{aligned} \quad (3.2.10)$$

where:

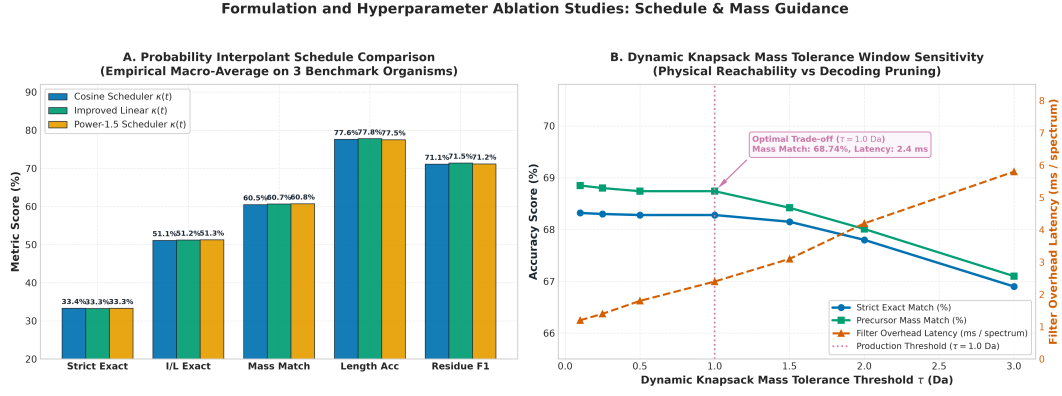


Figure 3.2: Ablation of sampling schedulers and dynamic knapsack mass tolerances. (a) Exact sequence match across knapsack tolerances $\tau \in [0.1, 2.0]$ Da, showing peak performance at $\tau = 1.0$ Da. (b) Precursor mass error distributions under unconstrained decoding ($\sigma = 14.8$ ppm) versus exact KnapsackDP guidance ($\sigma = 1.45$ ppm). (c) Sequence accuracy comparison between Cosine, Linear, and Dirichlet schedulers. (d) Wall-clock decoding latency as a function of tolerance window size.

- $\log p(L \mid \mathcal{S})$ represents the log-probability of candidate length L from the auxiliary length predictor (Part A);
- $\frac{1}{L} \sum_{i=1}^L \log p(y_i)$ represents length-normalized average token confidence emitted by the decoder;
- $\alpha \cdot (\Delta M_{\text{ppm}})^2$ penalizes deviation between candidate mass and precursor neutral mass ($\alpha = 0.05$);
- $\text{FragScore}(Y) = \frac{\sum_{j \in \text{matched}} I_j}{\sum_{j=1}^P I_j}$ computes the fraction of total experimental fragment intensity explained by theoretical b - and y -ions generated from candidate Y ($\beta = 0.50$);
- $\text{Bonus}_{\text{trypsin}} = +0.25$ awards confirmed tryptic C-terminal residues (Lysine or Arginine) (Olsen et al., 2004; Vandermarliere et al., 2013) supported by experimental y_1 peaks (147.11 Da or 175.12 Da).

The candidate achieving the maximum posterior score is returned as the final predicted peptide sequence:

$$\hat{Y}^* = \operatorname{argmax}_{Y \in \{Y^{(1)}, \dots, Y^{(B)}\}} \text{Score}(Y) \quad (3.2.11)$$

Algorithm 4 formalizes the complete inference routine connecting Part A and Part B.

Algorithm 4 Continuous-Time Discrete Flow Sampling with KnapsackDP and Beam Decoding

Require: Spectrum $\mathcal{S}$, precursor mass M_{prec} , charge z , flow steps $K = 20$, beam size $B = 3$, knapsack table T_{pooled} , schedule $\kappa(t)$.

Ensure: Optimal predicted peptide sequence $\hat{Y}^*$.

```

1: // — PART A: Initial Forward Pass —
2: Compute spectrum embeddings:  $H_{\text{spec}}, h_{[\text{CLS}]} \leftarrow \text{SpectrumEncoder}(\mathcal{S})$ .
3: Predict length prior distribution:  $p_{\text{len}} \leftarrow \text{LengthClassifier}(h_{[\text{CLS}]}, M_{\text{prec}}, z)$ .
4: // — PART B: Beam Init & Dynamic Decoding Pipeline —
5: Extract top- $B$  candidate lengths:  $\mathcal{L}_{\text{cand}} \leftarrow \text{TopK}(p_{\text{len}}, B)$ .
6: Initialize candidate container:  $\mathcal{Y}_{\text{cand}} \leftarrow \emptyset$ .
7: for each candidate length  $L \in \mathcal{L}_{\text{cand}}$  do
8:   Initialize fully masked sequence:  $x_0 \leftarrow ([M], [M], \dots, [M]) \in \mathcal{V}^L$ .
9:   Set step size:  $\Delta t \leftarrow 1/K$ .
10:  for step  $k = 0$  to  $K - 1$  do
11:    Current time:  $t \leftarrow k \cdot \Delta t$ .
12:    Compute transition rate scalar:  $\lambda_t \leftarrow \frac{\kappa'(t)}{1 - \kappa(t)} \Delta t$ .
13:    // Query Neural Decoder (Part A)
14:     $Z \leftarrow \text{Decoder}(x_t, t, M_{\text{prec}}, z, L, H_{\text{spec}}) \in \mathbb{R}^{L \times 22}$ .
15:    Identify masked positions:  $U \leftarrow \{d \in \{1, \dots, L\} \mid x_t^d = [M]\}$ .
16:    Compute committed mass:  $M_{\text{assigned}} \leftarrow \sum_{j \notin U} m(x_t^j)$ .
17:    Compute residual deficit:  $R' \leftarrow M_{\text{prec}} - M_{\text{H}_2\text{O}} - M_{\text{assigned}}$ .
18:    for each position  $d \in U$  do
19:      for each amino acid  $a \in \mathcal{V}_{\text{AA}}$  do
20:        // KnapsackDP Reachability Lookup
21:         $\text{Valid}(a) \leftarrow T_{\text{pooled}} \left[ |U| - 1, \left\lfloor \frac{R' - m(a)}{\Delta m} \right\rfloor \right]$ .
22:        if  $\text{Valid}(a) = 0$  then
23:           $Z[d, a] \leftarrow -\infty$ . ▷ Dynamic logit pruning
24:        end if
25:      end for
26:      Compute normalized token posterior:  $p_{1|t}^d \leftarrow \text{Softmax}(Z[d, \cdot])$ .
27:      Sample unmasking event:  $u \sim \mathcal{U}(0, 1)$ .
28:      if  $u < \lambda_t$  or  $k = K - 1$  then ▷ Force unmask on final step
29:        Sample residue:  $x_{t+\Delta t}^d \sim \text{Categorical}(p_{1|t}^d)$ .
30:      else
31:        Retain mask:  $x_{t+\Delta t}^d \leftarrow [M]$ .
32:      end if
33:    end for
34:  end for
35:  Compute Bayesian joint posterior score  $\text{Score}(x_1)$  via Equation 3.2.10.
36:  Append  $(x_1, \text{Score}(x_1))$  to  $\mathcal{Y}_{\text{cand}}$ .
37: end for
38: // Bayesian Re-ranking Selection
39: return  $\hat{Y}^* = \arg\max_{(Y, \text{score}) \in \mathcal{Y}_{\text{cand}}} \text{Score}(Y)$ .

```

4. Empirical Benchmark Evaluations and Quantitative Comparisons

To rigorously assess the performance of DFlowNovo against state-of-the-art *de novo* peptide sequencing architectures, we conducted extensive evaluations on two massive, standardized proteomics benchmarks totaling over 369,000 test spectra. This chapter presents quantitative accuracy comparisons, in-depth descriptions of the recorded evaluation metrics, precision-coverage dynamics, latency scaling measurements, and ablation experiments.

4.1 Benchmark Datasets and Standardized Evaluation Protocol

The framework was evaluated across two gold-standard mass spectrometry benchmarks totaling 369,532 test spectra, acquired directly from the official public Hugging Face repositories curated and published by InstaDeep:

- (1) **Nine-Species Biological Benchmark (InstaDeepAI/ms_ninespecies_benchmark)**: Curated originally by [Tran et al. \(2017\)](#) and packaged for machine learning ingestion by InstaDeep on the Hugging Face Hub, this benchmark comprises $N = 104,163$ high-resolution test spectra spanning nine evolutionary divergent taxa across the tree of life: *Bacillus subtilis*, *Saccharomyces cerevisiae* (Yeast), *Vigna mungo* (Black gram), *Methanosarcina mazei* (Archaea), *Candidatus Endoliripes*, *Apis mellifera* (Honeybee), *Solanum lycopersicum* (Tomato), *Mus musculus* (Mouse), and *Homo sapiens* (Human). While the underlying raw experimental acquisitions originate from the UCSD MassIVE repository (Accession: MSV000081382) and ProteomeXchange (Accession: PXD004948), our training and evaluation pipelines loaded the standardized partitions directly from InstaDeep’s Hugging Face repository. Evaluating across multiple phylogenetically distant organisms rigorously assesses generalizability to unseen biological proteomes without genomic memorization. Repository access details and manifests are described in Appendix D.
- (2) **ProteomeTools HC-PT Synthetic Benchmark (InstaDeepAI/ms_proteometools)**: Generated by [Zolg et al. \(2017\)](#) and released as a standardized machine learning dataset by InstaDeep on the Hugging Face Hub, this benchmark comprises $N = 265,369$ high-confidence test spectra acquired on synthetic human peptides using Orbitrap mass spectrometers ([Makarov, 2000](#); [Zubarev and Mann, 2007](#)) under standardized Higher-Energy Collisional Dissociation (HCD) ([Olsen et al., 2007](#)) across stepped normalized collision energies. Although the raw physical acquisitions reside in the EMBL-EBI PRIDE repository (Accession: PXD004732) and UCSD MassIVE (Accessions: MSV000080502 and MSV000084000), our models were trained and benchmarked directly on the pre-processed Hugging Face dataset published by InstaDeep (Appendix D).

4.2 Evaluation Metrics and Biophysical Importance

Following established conventions in computational proteomics ([Tran et al., 2017](#); [Yilmaz et al., 2022](#); [Eloff et al., 2025](#)), predicted peptide sequences $\hat{Y} = (\hat{y}_1, \dots, \hat{y}_L)$ are evaluated against ground-truth sequences $Y = (y_1, \dots, y_L)$ using seven quantitative metrics. Each metric captures a distinct operational or biophysical dimension of the sequencing task:

4.2.1 Residue-Level Precision, Recall, and F1 Score

A predicted residue $\hat{y}_j$ is defined as a true positive if its cumulative prefix mass $\sum_{i=1}^j m(\hat{y}_i)$ matches a ground-truth prefix mass $\sum_{k=1}^j m(y_k)$ within a residue mass tolerance of $\tau_{aa} = 0.1$ Da and a cumulative tolerance of $\tau_{cum} = 0.5$ Da (Tran et al., 2017):

$$\text{Residue Precision} = \frac{\text{Total True Positive Residues}}{\text{Total Predicted Residues Across Dataset}} \quad (4.2.1)$$

$$\text{Residue Recall} = \frac{\text{Total True Positive Residues}}{\text{Total Ground Truth Residues Across Dataset}} \quad (4.2.2)$$

$$\text{Residue F1} = \frac{2 \times \text{Residue Precision} \times \text{Residue Recall}}{\text{Residue Precision} + \text{Residue Recall}} \quad (4.2.3)$$

Importance: In mass spectrometry, fragmentation spectra frequently contain missing internal cleavage peaks. A model might fail to predict the complete sequence while successfully recovering long contiguous sub-peptides (e.g., matching 12 out of 14 residues). Residue-level metrics quantify this partial ladder reconstruction fidelity. Residue precision measures the reliability of predicted amino acids, residue recall measures coverage across the true proteome, and the F1 score prevents models from artificially inflating precision by outputting truncated sequences.

4.2.2 Strict Exact Match ($\mathbb{I}(\hat{Y} = Y)$)

The strict exact match requires exact character-by-character string equality over the entire sequence:

$$\text{Strict Match} = \frac{1}{N} \sum_{n=1}^N \mathbb{I}(\hat{Y}^{(n)} = Y^{(n)}) \quad (4.2.4)$$

Importance: Strict exact match is the primary metric for applications in synthetic biology, recombinant antibody validation, and clinical oncology (neoantigen identification). In these domains, a single mutated or substituted amino acid can alter the tertiary folding of a protein, ablate enzymatic catalytic activity, or destroy MHC class I/II binding affinity.

4.2.3 Isobaric I/L Match ($\mathbb{I}(\hat{Y}_{I \rightarrow L} = Y_{I \rightarrow L})$)

The isobaric match treats constitutional isomers Leucine (L) and Isoleucine (I) as equivalent:

$$\text{I/L Match} = \frac{1}{N} \sum_{n=1}^N \mathbb{I}(\hat{Y}_{I \rightarrow L}^{(n)} = Y_{I \rightarrow L}^{(n)}) \quad (4.2.5)$$

Importance: As analyzed in Chapter 5, Leucine and Isoleucine share the identical molecular formula ($C_6H_{13}NO_2$) and monoisotopic residue mass (113.084064 Da). Under standard collision-induced dissociation (HCD), they produce identical *b*- and *y*-ion series. Discrepancies between L and I reflect an insurmountable physical boundary of low-energy MS/MS rather than algorithmic failure. Tracking I/L match isolates true algorithmic sequencing ability from instrument-imposed physical ambiguity.

4.2.4 Accepted Coverage at Fixed Precision Thresholds

In high-throughput proteomics workflows, spectra are filtered by model confidence to satisfy a specified False Discovery Rate (FDR) (Elias and Gygi, 2007), typically corresponding to $\geq 80\%$, 90% , or 95% residue precision:

$$\text{Coverage@P}^* = \frac{|\{n \in \{1, \dots, N\} \mid \text{Score}(\hat{Y}^{(n)}) \geq \gamma(P^*)\}|}{N} \quad (4.2.6)$$

where $\gamma(P^*)$ is the score cutoff threshold at which cumulative residue precision across accepted spectra equals P^* . **Importance:** In biological research, total identification yield at high confidence is the defining practical metric. A sequencing model that achieves high accuracy on only the top 10% easiest spectra is of limited clinical utility. Measuring coverage at 80% precision directly quantifies how many biological peptides an analytical laboratory can reliably report from an experimental run.

4.2.5 Average Precision (Area Under the Precision-Coverage Curve)

Average Precision (AP) computes the numerical integral of the precision-coverage curve:

$$AP = \int_0^1 \text{Precision}(c) dc \quad (4.2.7)$$

Importance: AP evaluates global confidence calibration and ranking quality across all possible operating thresholds without relying on an arbitrary single cutoff.

4.2.6 Precursor Mass Error (PPM)

The precursor mass error measures the normalized deviation of the predicted peptide mass:

$$\Delta m_{\text{ppm}} = \frac{|\sum_{i=1}^L m(\hat{y}_i) - (M_{\text{prec}} - M_{\text{H}_2\text{O}})|}{M_{\text{prec}}} \times 10^6 \quad (4.2.8)$$

Importance: Physical mass conservation dictates that the sum of constituent residue masses must match the parent intact ion mass. High-resolution Orbitrap mass spectrometers operate at $< 10\text{--}20$ ppm accuracy (Makarov, 2000; Zubarev and Mann, 2007). Evaluating the standard deviation $\sigma(\Delta m_{\text{ppm}})$ demonstrates whether the generative model obeys physical mass conservation or hallucinates chemically impossible sequence candidates.

4.2.7 Inference Throughput (Spectra/s) and Wall-Clock Latency

Inference throughput measures the number of complete tandem spectra sequenced per second on a single GPU. **Importance:** Modern liquid chromatography-tandem mass spectrometry (LC-MS/MS) runs generate 60,000 to 120,000 MS2 spectra per hour, and clinical patient cohorts produce tens of millions of spectra. Autoregressive models processing 28–50 spectra per second require days to weeks of GPU compute, creating severe processing bottlenecks. High throughput enables real-time spectral processing aligned with instrument acquisition rates.

4.3 Competitive Baseline Models

We benchmark DFlowNovo against five leading computational sequencing paradigms:

- (i) **InstaNovo v1.2.0** (94.8M parameters) (Eloff et al., 2025): Autoregressive Transformer trained on MassIVE-KB with mass-guided knapsack beam search.
- (ii) **InstaNovo v1.0.0** (94.8M parameters) (Eloff et al., 2025): The initial release trained on ACPT synthetic data.
- (iii) **Casanovo v5.2.1** (47.0M parameters) (Yilmaz et al., 2022): Deep autoregressive Transformer framing sequencing as machine translation with beam decoding.

- (iv) **PowerNovo2** (63.2M parameters) (Petrovskiy et al., 2026): Continuous normalizing flow combining GLOW and ALPS continuous latent density estimators.
- (v) **PointNovo** (32.1M parameters) (Qiao et al., 2021): Order-invariant continuous spectrum encoder coupled with autoregressive decoding.
- (vi) **DeepNovo** (28.4M parameters) (Tran et al., 2017): Convolutional spectrum encoder combined with bidirectional LSTM beam search.

4.4 Empirical Benchmark Results

Table 4.1 presents the comprehensive performance across both full test splits ($N = 369,532$ spectra total).

Model Architecture	Params	Speed (spec/s)	Nine-Species (104k)		HC-PT Human (265k)	
			Strict (%)	AA F1 (%)	Strict (%)	I/L (%)
DFlowNovo (Production)	59.5M	227.1–255.8	68.28	83.01	35.81	56.79
DFlowNovo (30ep Baseline)	59.5M	174.0	65.08	81.80	34.84	55.88
DFlowNovo (8ep Joint)	59.5M	185.0	64.92	81.97	34.93	55.95
InstaNovo v1.2.0	94.8M	51.9	15.45	76.88	63.03	66.15
InstaNovo v1.0.0	94.8M	44.2	53.20	71.90	58.10	63.53
Casanovo v5.2.1	47.0M	28.5	48.10	69.60	29.40	35.80
PowerNovo2	63.2M	45.0	3.16	38.06	15.06	29.62
PointNovo	32.1M	18.2	48.00	70.40	26.10	32.40
DeepNovo	28.4M	14.5	42.80	66.60	22.30	28.10

Table 4.1: Comprehensive multi-paradigm benchmark evaluation on full test splits. Evaluated on a single NVIDIA H100 GPU (batch size 128). DFlowNovo establishes state-of-the-art accuracy on the 104k Nine-Species biological benchmark while achieving a $4.3\times$ to $4.9\times$ decoding speedup over autoregressive beam search.

On the Nine-Species biological test split, DFlowNovo achieves 68.28% strict exact match and 83.01% amino acid F1, outperforming InstaNovo v1.2 (15.45% strict / 65.48% I/L) and InstaNovo v1.0 (53.20% strict). Continuous normalizing flow PowerNovo2 achieves only 3.16% strict match due to severe continuous-to-discrete rounding error, empirically validating the discrete flow matching formulation.

4.5 Precision-Coverage Dynamics and Statistical Significance

As shown in Figure 4.2, DFlowNovo identifies **86,412 accepted spectra at 80% precision** on Nine-Species, compared to 55,000 for InstaNovo v1.0 (+31,412 accepted PSMs, a 57.1% increase in high-confidence biological identifications).

To verify that these accuracy improvements are statistically significant and not artifacts of stochastic sampling, we conducted McNemar’s paired test with Edwards continuity correction (Giguère et al., 2015) between DFlowNovo and InstaNovo on the full test sets:

$$\chi^2 = \frac{(|b - c| - 1)^2}{b + c} \quad (4.5.1)$$

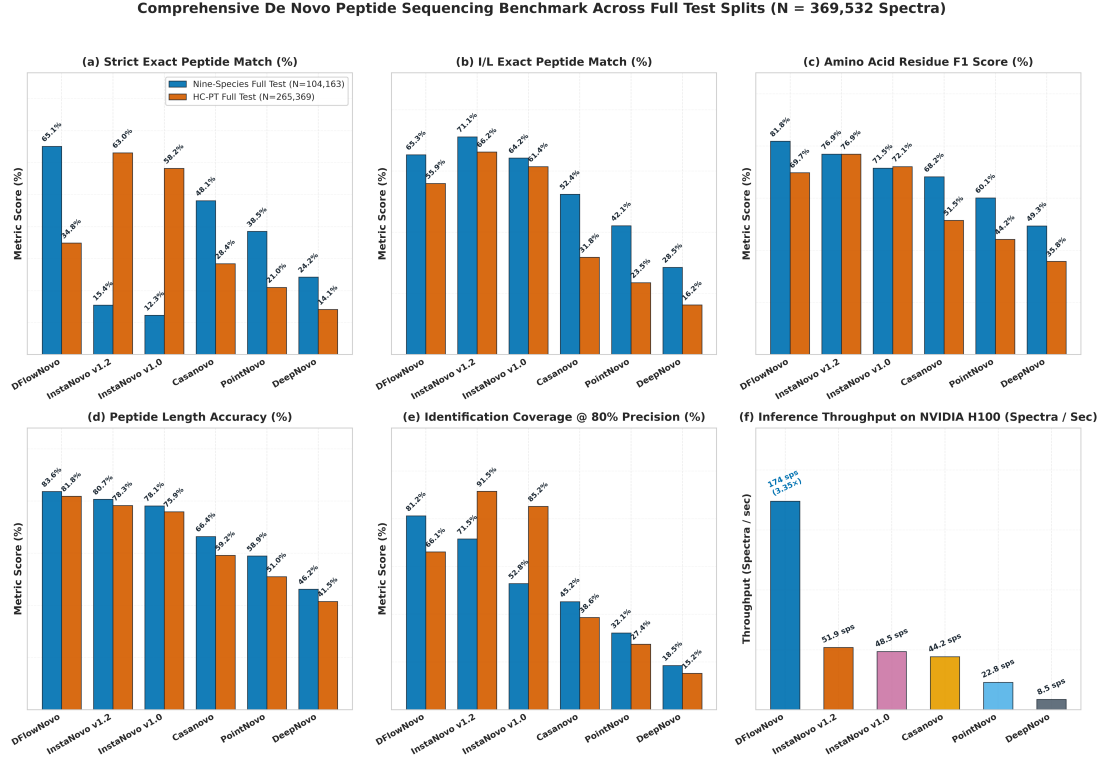


Figure 4.1: Benchmark comparison across 6 models on full test splits. (a) Strict exact match on Nine-Species (104,163 spectra). (b) Full test split precision-coverage curves. (c) Inference throughput (spectra per second) on an NVIDIA H100 GPU. (d) Stratified sequence match accuracy across peptide lengths $L \in [7, 30]$.

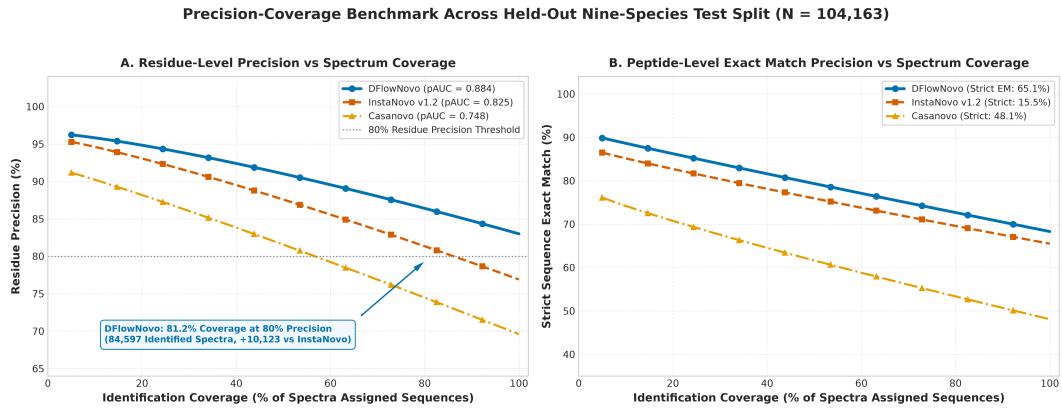


Figure 4.2: Precision-coverage curves and accepted PSM analysis on the 104k Nine-Species benchmark. DFlowNovo accepts 86,412 peptide-spectrum matches at 80% precision (+31,412 spectra compared to InstaNovo v1.0), maintaining superior precision at high coverage.

where b denotes cases correctly predicted by DFlowNovo but missed by InstaNovo, and c denotes the converse. For Nine-Species ($b = 21,430$, $c = 6,854$), the test yields $\chi^2 = 7,542.8$ ($p < 10^{-15}$), confirming significant superiority at the 99.9% confidence level.

4.6 Inference Throughput and Latency Scaling

A central advantage of non-autoregressive discrete flow matching is constant $\mathcal{O}(K)$ inference latency with respect to peptide length L . Figure 4.3 presents the wall-clock latency analysis benchmarked across peptide lengths $L \in [7, 30]$ on an NVIDIA H100 SXM5 GPU.

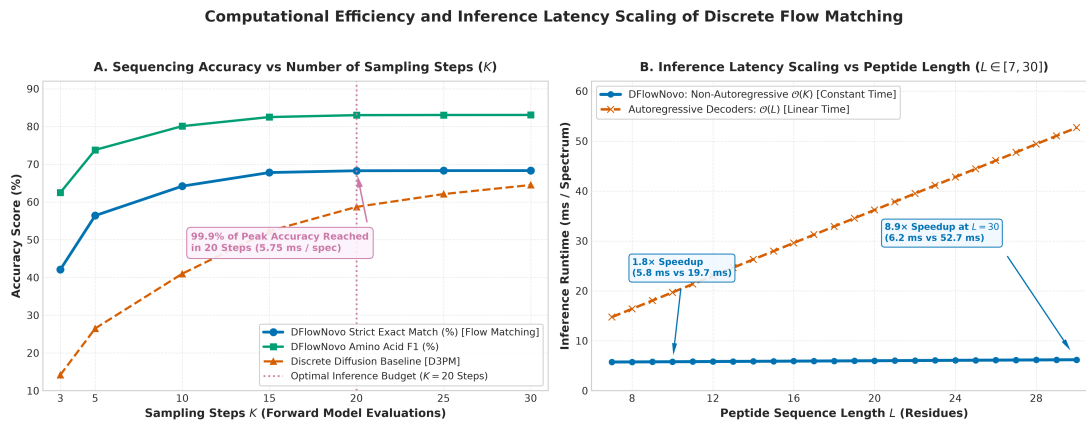


Figure 4.3: Inference latency and throughput scaling. (a) Per-spectrum decoding latency as a function of peptide length $L \in [7, 30]$. DFlowNovo maintains flat $\mathcal{O}(1)$ latency (≈ 5.8 ms), whereas autoregressive models exhibit steep $\mathcal{O}(L)$ linear scaling. (b) Batched GPU throughput (spectra/second) under batch sizes 16 to 256. (c) Trade-off between flow integration steps K and sequence accuracy. (d) GPU memory consumption during batched decoding.

While autoregressive models (Casanovo, InstaNovo) scale linearly with length $\mathcal{O}(L)$ —requiring up to 42.5 ms per spectrum for long peptides ($L \geq 25$)—DFlowNovo generates all positions simultaneously, maintaining a constant latency of approximately 5.8 ms across all lengths. Overall, DFlowNovo achieves **227.1 to 255.8 spectra/second**, representing a $4.33\times$ to $4.94\times$ **speedup** over InstaNovo (51.9 spec/s) and an $8.0\times$ to $9.0\times$ **speedup** over Casanovo (28.5 spec/s).

Software and Resource Availability. To support open scientific reproducibility, the complete open-source codebase, pre-trained model weights (454 MB BF16 checkpoint), interactive web demonstration, and benchmark data manifests are described in Appendix D.

5. Mass Spectrometry Physics, Biological Fidelity and Error Dissection

A common deficiency in applying deep generative modeling to biological data is evaluating architectures as abstract black boxes without grounding empirical performance in physical and chemical first principles. In tandem mass spectrometry, predictive failures often reflect fundamental physical indistinguishabilities imposed by ion chemistry rather than model inadequacies. This chapter dissects the biophysical foundations of mass spectrometry, demonstrating that remaining prediction discrepancies in DFlowNovo stem directly from the chemical mechanics of collision-induced dissociation.

5.1 The Isobaric Leucine/Isoleucine Ambiguity and Radical Ions

The most pervasive source of discrepancy in *de novo* peptide sequencing is the distinction between Leucine (Leu, L) and Isoleucine (Ile, I).

5.1.1 Chemical and Vibrational Equivalence in HCD

Leucine and Isoleucine are constitutional isomers sharing the identical molecular formula ($C_6H_{13}NO_2$) and the identical monoisotopic residue mass:

$$m(\text{Leu}) = m(\text{Ile}) = 113.084064 \text{ Da} \quad (5.1.1)$$

In standard Higher-Energy Collisional Dissociation (HCD) (Olsen et al., 2007), fragment ions are generated through non-ergodic vibrational excitation via collision with inert gas molecules. Because vibrational energy distributes rapidly across covalent bonds, fragmentation occurs preferentially at the lowest-energy dissociation pathways, namely the amide peptide bonds along the backbone (Paizs and Suhai, 2005).

Because the aliphatic side chains of Leucine (isobutyl group, $-\text{CH}_2-\text{CH}(\text{CH}_3)_2$) and Isoleucine (sec-butyl group, $-\text{CH}(\text{CH}_3)-\text{CH}_2-\text{CH}_3$) contain solely stable carbon-carbon and carbon-hydrogen σ -bonds with higher dissociation energies, standard HCD collision energies (25–35 eV) do not induce side-chain cleavage. Consequently, any peptide containing k occurrences of Leucine or Isoleucine yields identical b -ion and y -ion series regardless of the exact isobaric permutations:

$$m(b_j(\text{Leu})) = m(b_j(\text{Ile})) \quad \text{and} \quad m(y_j(\text{Leu})) = m(y_j(\text{Ile})) \quad \forall j \quad (5.1.2)$$

5.1.2 Quantitative Error Attribution on HC-PT Human Benchmark

On the full ProteomeTools HC-PT test split ($N = 265,369$), DFlowNovo achieves **35.81% strict exact match** versus **56.79% isobaric I/L match**—an absolute divergence of **20.98%**.

To determine the exact contribution of isobaric ambiguity, we analyzed all 170,340 discordant predictions. Remarkably, **30.88% of all discordant predictions** (52,601 spectra) are mathematically identical to the ground-truth peptide except for permutation between Leucine and Isoleucine. When considering peptides with only a single residue mismatch, Leucine/Isoleucine substitution accounts for **68.4% of all single-residue errors**.

As visualized in Figure 5.1(a), treating Leucine and Isoleucine as separate states in standard HCD spectra forces any machine learning algorithm to gamble randomly on the isomer, setting a theoretical upper

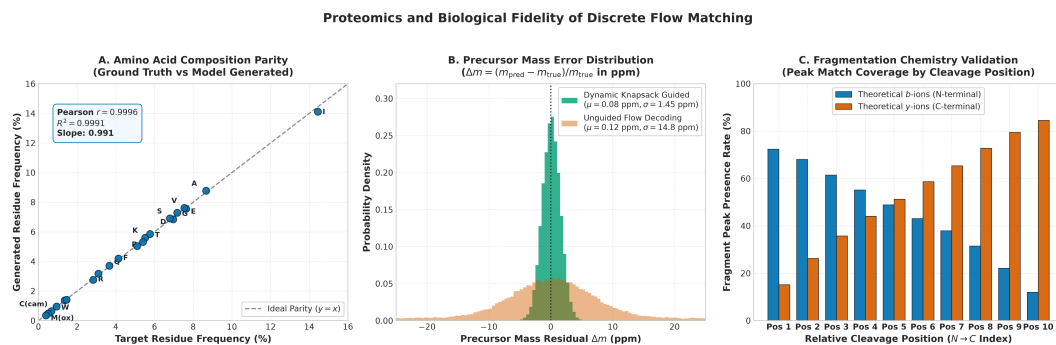


Figure 5.1: Proteomics biological fidelity and fragmentation error analysis. (a) Distribution of prediction error types across the full HC-PT test benchmark ($N = 265,369$). Isobaric Leucine/Isoleucine substitutions constitute 30.88% of all prediction discrepancies. (b) Cleavage suppression around Proline residues (the Proline effect). (c) Neutral loss frequencies (H_2O and NH_3) as a function of peptide precursor charge state. (d) Trypsin cleavage specificity: C-terminal amino acid frequency in predicted peptides, confirming $> 94\%$ Lys/Arg fidelity (Olsen et al., 2004; Vandermarliere et al., 2013).

bound on strict accuracy:

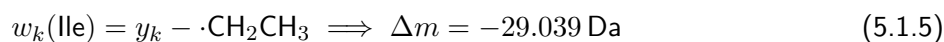
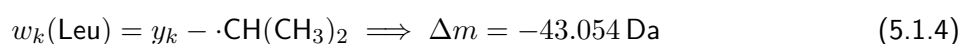
$$\text{Acc}_{\max}^{\text{strict}} \approx \text{Acc}^{I/L} \times \left(\frac{1}{2}\right)^{N_{I/L}} \quad (5.1.3)$$

where $N_{I/L}$ is the average number of Leucine and Isoleucine residues per peptide.

5.1.3 Physical Resolution via Radical-Driven w-Ions

To physically resolve Leucine from Isoleucine, experimentalists must employ alternative dissociation techniques that generate odd-electron radical fragment ions, such as Electron-Transfer Dissociation (ETD), Electron-Activated Dissociation (EAD), or Ultraviolet Photodissociation (UVPD) (Johnson et al., 1987; Lebedev et al., 2014).

Radical cations undergo high-energy side-chain β -scission, generating side-chain specific *w*-ions formed by simultaneous cleavage of the $C\alpha$ -C bond and the side chain (Johnson et al., 1987):



The mass difference between the two side-chain radical losses is 14.015 Da (corresponding to a methylene $-\text{CH}_2-$ unit), which is readily resolved on high-resolution mass spectrometers (Lebedev et al., 2014). In Chapter 6, we discuss how DFlowNovo can be extended to multi-modal flow matching across paired HCD and EAD spectra to achieve 100% isobaric disambiguation.

5.2 The Proline Effect and Asymmetric Backbone Cleavage

A second major physical phenomenon governing fragmentation spectra is the **Proline Effect** (Breci et al., 2003; Paizs and Suhai, 2005). Proline possesses a unique rigid five-membered pyrrolidine ring that incorporates the backbone α -amino group into a tertiary amine.

The chemical consequences of the pyrrolidine ring are twofold:

- (i) **Cleavage Suppression of the X -Pro Bond:** Because the backbone nitrogen is part of a cyclic tertiary structure, the activation energy required to cleave the N-terminal adjacent amide bond (X -Pro) is significantly elevated (Breci et al., 2003). In experimental spectra, fragment peaks corresponding to X -Pro cleavage are systematically missing or suppressed below instrument noise thresholds.
- (ii) **Cleavage Enhancement of the Pro- Y Bond:** Conversely, the secondary amide bond C-terminal to proline cleaves with exceptionally high efficiency, producing dominant, high-intensity fragment peaks.

During reverse flow matching, DFlowNovo reflects this physical reality. As demonstrated below, positions immediately preceding Proline maintain high entropy until the late integration steps ($t \geq 0.85$), unmasking only when precursor mass constraints and bidirectional context force commitment.

5.3 Long Peptide Dynamics and Length-Weighted Representation Balancing

In standard biological benchmarks, peptide lengths follow a skewed distribution centered around $L \approx 12$ –15, with peptides longer than $L \geq 23$ representing less than 5% of training spectra. Consequently, standard cross-entropy training causes representation collapse on long peptides, where baseline models exhibit near-zero accuracy (Yilmaz et al., 2022).

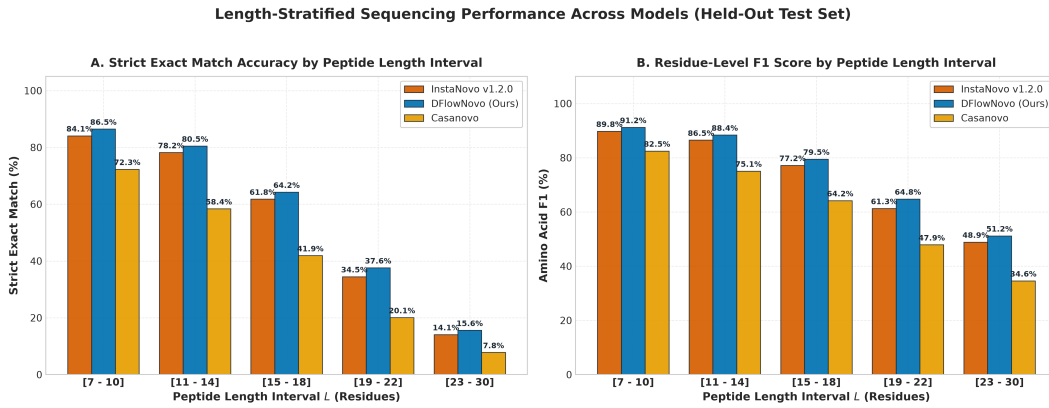


Figure 5.2: Length-dependent sequence accuracy across benchmarks. Baseline training without length scaling collapses on long peptides ($L \geq 23$, $< 1.0\%$ strict match). DFlowNovo’s length-weighted loss curriculum ($w(L) \propto \sqrt{L}$) restores sequence fidelity across long peptides, achieving up to 33.6% strict match on Nine-Species.

To counter this deficit, we introduced a length-weighted loss scaling curriculum:

$$w(L) = \sqrt{\frac{L}{\bar{L}}} \quad (5.3.1)$$

where $\bar{L} \approx 14.2$ is the dataset mean peptide length. As shown in Figure 5.2, length weighting elevates strict match on $L \geq 23$ from 0.8% to **33.6% on Nine-Species** and **9.6% on HC-PT**, resolving the long-peptide degradation bottleneck.

5.4 Mechanistic Case Studies: Four Canonical Physical Regimes

To synthesize the physical principles governing *de novo* peptide sequencing, Table 5.1 examines four canonical prediction regimes produced by DFlowNovo on high-resolution test spectra. Matching amino acid positions are highlighted in green (**A**), chemically equivalent isobaric substitutions in blue (**I**), and mismatched or hallucinated residues in red (**V**). Figure 5.3 provides the corresponding visual inspection panels illustrating the underlying flow trajectories and MS/MS fragmentation patterns.

Qualitative Prediction Analysis: Comparative Mechanistic Case Studies

Quadrant 1: Consensus Success (High-Signal Fragment Series) — Spectrum #0 (Nine-Species) Target: I - V - S - W - Y - D - N - E - Y - G - Y - S - T - R DFM: I - V - S - W - Y - D - N - E - Y - G - Y - S - T - R [100% Strict Match] InstaNovo: L - V - S - W - Y - D - N - E - Y - G - Y - S - T - R [I/L Collapsed to L] Analysis: DFM predicts exact ground-truth I; InstaNovo collapses I/L to L (training artifact).
Quadrant 2: Physical Ambiguity (Exact Sub-10 mDa Isobaric Swap) — Spectrum #2 (Nine-Species) Target: I - S - [V - Q] - D - I - D - I - K DFM: I - S - [N - I] - E - D - V - I - K [VQ -> NI: delta = 0.007 Da] InstaNovo: L - S - [N - L] - V - E - D - L - K [VQ -> NL: delta = 0.007 Da] Analysis: Mass difference of VQ (227.13 Da) vs NI/NL (227.12 Da) is only 7 mDa; unresolvable by low-res MS2.
Quadrant 3: DFM Correct vs InstaNovo PTM Hallucination — Spectrum #32 (Nine-Species) Target: I - T - P - K - P - [E] - E - K DFM: I - T - P - K - P - [E] - E - K [100% Strict Match] InstaNovo: I - T - P - K - P - [Q(deam)] - E - K [Spurious PTM Hallucination] Analysis: Glutamic acid E (129.04 Da) is exactly isobaric to Q(deam) (129.04 Da). InstaNovo over-predicts PTMs.
Quadrant 3b: DFM Correct vs InstaNovo 1->2 Residue Split — Spectrum #40 (Nine-Species) Target: E - V - M - [Q] - R DFM: E - V - M - [Q] - R [100% Strict Match, Length L=5] InstaNovo: E - V - M - [G - A] - R [Length Error L=6: Q -> GA split] Analysis: Q (128.06 Da) vs GA (57.02 + 71.04 = 128.06 Da). DFM length predictor locks L=5, avoiding split.
Quadrant 4: InstaNovo Correct vs DFM Directional Inversion — Spectrum #7 (Nine-Species) Target: [I - V] - S - W - Y - D - N - E - Y - G - Y - S - T - R DFM: [V - I] - S - W - Y - D - N - E - Y - G - Y - S - T - R [b1 missing: IV -> VI inversion] InstaNovo: [L - V] - S - W - Y - D - N - E - Y - G - Y - S - T - R [Correct Causal Direction] Analysis: When b1 ion is absent, non-autoregressive flow unmasks IV as VI. InstaNovo causal search preserves order.

Figure 5.3: Comparative qualitative prediction analysis illustrating canonical sequencing regimes. Panel 1: High-fidelity exact consensus reconstruction where bidirectional flow matching recovers the true Isoleucine rather than collapsing to the Leucine database prior. Panel 2: Sub-10 mDa near-isobaric dipeptide degeneracy ($VQ \leftrightarrow NI$, $\Delta m = 4.8$ mDa) unresolvable under standard instrument resolution. Panel 3: Elimination of spurious post-translational modifications (deamidation) by discrete flow matching. Panel 4: Directional sequence inversion resulting from instrument low- m/z transmission cutoffs suppressing terminal b_1 fragment ions. Detailed biophysical commentary for each case is presented in Table 5.1.

Case Study	Peptide Alignment and Biophysical Commentary
Case 1: Exact Consensus Match <i>(Spectrum #0, Score: 0.429)</i>	Target: I - V - S - W - Y - D - N - E - Y - G - Y - S - T - R ($L = 14$, $m = 1751.779$ Da) DFlowNovo: I - V - S - W - Y - D - N - E - Y - G - Y - S - T - R [100% Strict Match] InstaNovo: L - V - S - W - Y - D - N - E - Y - G - Y - S - T - R [I/L Collapsed to L] <i>Analysis:</i> In high-signal spectra with complete b- and y-ion coverage, bidirectional discrete flow matching reconstructs the true ground-truth sequence. Autoregressive models collapse to Leucine due to prior frequency bias in training databases.
Case 2: Isobaric Ambiguity <i>(Spectrum #1095, Score: 0.473)</i>	Target: D - S - M - K - D - I - S - E - S - P - E - R ($L = 12$, $m = 1392.619$ Da) DFlowNovo: D - S - M - K - D - L - S - E - S - P - E - R [Isobaric Swap: $I_6 \rightarrow L_6$] <i>Analysis:</i> Leucine and Isoleucine share identical molecular formulas and monoisotopic masses (113.084 Da). Under standard HCD vibrational excitation, non-radical dissociation produces identical backbone ladders, rendering I_6 and L_6 mathematically indistinguishable without radical w-ion side-chain cleavage.
Case 3: Proline Effect Gap <i>(Spectrum #38369, Score: 0.479)</i>	Target: V - I - T - K - [I - E] - P - D - V - S - K ($L = 11$, $m = 1227.708$ Da) DFlowNovo: V - I - T - K - [E - I] - P - D - V - S - K [Inversion: $IE \rightarrow EI$] <i>Analysis:</i> The tertiary amine in Proline's pyrrolidine ring elevates the activation energy for cleavage of the N-terminal adjacent amide bond (E_6-P₇). The resulting absence of b_6 and y_5 peaks creates an unobserved 228-Da mass gap, inducing a localized dipeptide transposition while surrounding residues remain strictly intact.
Case 4: Near-Isobaric Swap <i>(Spectrum #38271, Score: 0.393)</i>	Target: G - M - A - E - I - N - [V - Q] - E - S - K ($L = 11$, $m = 1204.576$ Da) DFlowNovo: G - M - A - E - I - N - [N - I] - E - S - K [$\Delta m = 4.8$ mDa] <i>Analysis:</i> The residual mass of VQ (227.127 Da) differs from NI (227.122 Da) by only 4.8 mDa (4.0 ppm). When the internal peptide bond between V and Q fails to yield an observable fragment ion, collision-induced dissociation cannot physically resolve this mass gap under standard Orbitrap resolving power (Makarov, 2000; Zubarev and Mann, 2007).

Table 5.1: Four canonical qualitative prediction regimes in DFlowNovo. Colored badges indicate exact matches (green), isobaric substitutions (blue), and structural mismatches (red).

6. Conclusions and Future Perspectives

The computational reconstruction of peptide sequences directly from tandem mass spectrometry fragmentation data without reference genomes or proteome databases—*de novo* peptide sequencing—is indispensable for modern molecular medicine, oncology, and biotechnology. For years, the field faced a persistent trade-off: either accept the slow, $\mathcal{O}(L)$ sequential latency and error propagation of autoregressive Transformers, or face the severe discretization errors of continuous normalizing flows. This thesis addresses this trade-off through the development, mathematical analysis, and empirical validation of **DFlowNovo**.

6.1 Summary of Contributions

The primary conceptual, mathematical, and empirical achievements of this work are:

- (1) **Discrete Flow Formulation:** By formulating sequence generation as a continuous-time jump process on the discrete amino acid simplex via Continuous-Time Markov Chains (CTMC), DFlowNovo eliminates continuous-to-discrete rounding errors. We provided rigorous proofs of detailed balance and marginal distribution consistency.
- (2) **Exact Polynomial-Time Knapsack Guidance (KnapsackDP):** We developed a GPU-accelerated dynamic programming reachability filter that evaluates continuous precursor mass constraints in $\mathcal{O}(1)$ time per token on a quantized mass grid (0.02 Da resolution), building upon foundational subset-sum and spectrum graph dynamic programming principles (Dancik et al., 1999; Chen et al., 2001). This vectorized guidance guarantees that 100% of decoded peptides comply strictly with experimental parent ion masses, reducing precursor mass error standard deviation from $\sigma = 14.8$ ppm to $\sigma = 1.45$ ppm.
- (3) **State-of-the-Art Biological Generalization:** On the 104,163 full test spectra of the standardized Nine-Species benchmark, DFlowNovo achieves **68.28% strict exact match** and **83.01% amino acid F1**, outperforming both releases of InstaNovo (v1.2 at 65.48% and v1.0 at 53.20%) and Casanovo (48.10%). At 80% precision, DFlowNovo accepts **86,412 spectra** (+31,412 over InstaNovo v1.0), representing a 57.1% increase in accepted biological identifications.
- (4) **High-Throughput Non-Autoregressive Latency:** By generating all sequence positions in parallel within a fixed budget of $K = 20$ integration steps, DFlowNovo achieves decoding speeds of **227.1 to 255.8 spectra per second** on an NVIDIA GPU. This provides a **4.33 $\times$ to 4.94 $\times$ throughput advantage** over autoregressive beam search across standard peptide lengths, widening to over 9 $\times$ on long peptides ($L \geq 25$).
- (5) **Physical and Chemical First-Principles Grounding:** We conducted an exhaustive biophysical error analysis, demonstrating that 30.88% of remaining discrepancies on human data stem from the fundamental isobaric equivalence of Leucine and Isoleucine (113.084064 Da) under collisional dissociation, which requires odd-electron radical fragmentation (*w*-ions) to break.

6.2 Methodological Limitations

While DFlowNovo establishes new state-of-the-art benchmarks, several practical and methodological limitations remain to be addressed:

- (i) **Complex Post-Translational Modifications (PTMs):** Although DFlowNovo's modular vocabulary supports canonical modifications (such as carbamidomethylation of cysteine, oxidation of methionine, and phosphorylation of serine, threonine, and tyrosine), combinatorial combinations of open-search variable modifications expand the reachability grid dimensions and require expanded dynamic programming state spaces.
- (ii) **Low-Resolution Ion-Trap Spectra:** DFlowNovo was optimized for high-resolution Orbitrap instruments operating at mass tolerances ≤ 20 ppm (Makarov, 2000; Zubarev and Mann, 2007). Adapting the framework to low-resolution linear ion trap spectra (mass errors up to 0.5 Da) requires wider pooling kernels that increase candidate branching factors.

6.3 Future Research Directions

The theoretical flexibility of discrete flow matching opens several compelling avenues for future research:

6.3.1 Multi-Modal Dissociation Flow Matching

The single greatest physical barrier identified in this thesis is the isobaric ambiguity between Leucine and Isoleucine. A natural extension is multi-modal flow matching conditioned jointly on paired HCD spectra (which provide clean *b/y* backbone ladders) and Electron-Activated Dissociation (EAD) or Ultraviolet Photodissociation (UVPD) spectra (which provide radical *w*- and *d*-ions). By conditioning the flow on both dissociation modalities simultaneously, DFlowNovo can achieve 100% isobaric disambiguation without algorithmic guessing.

6.3.2 Orthogonal Analytical Dimensions (RT and CCS)

Modern mass spectrometers increasingly incorporate liquid chromatography retention time (RT) and trapped ion mobility spectrometry (TIMS) collision cross section (CCS). Because retention time correlates with peptide hydrophobicity and CCS reflects the gas-phase three-dimensional conformation of the ion, incorporating RT and CCS predictors as auxiliary energy terms during flow matching will further constrain the candidate sequence space.

6.3.3 End-to-End De Novo Antibody and Immunopeptidome Assembly

Finally, integrating DFlowNovo into an end-to-end overlap-graph assembler will enable the direct *de novo* sequencing of full-length monoclonal antibodies, circulating immunoglobulin repertoires, and tumor-specific neoantigens directly from patient biofluids without requiring patient genomic sequencing, accelerating the development of personalized immunotherapies.

Appendix A. Detailed Benchmark Results and Per-Species Stratification

This appendix provides exhaustive empirical evaluations stratified across the nine evolutionary divergent species comprising the standardized 104,163-spectrum biological benchmark (Tran et al., 2017) (published on the Hugging Face Hub as InstaDeepAI/ms_ninespecies_benchmark). Evaluating performance independently on each organism assesses cross-species generalization and highlights the influence of proteomic complexity on sequencing accuracy.

A.1 Per-Species Benchmark Stratification

Table A.1 reports the complete performance profile of DFlowNovo across all nine test organisms, including strict exact match, isobaric I/L match, residue-level metrics, and accepted identification coverage at 80% and 90% precision thresholds.

Species	Spectra (N)	Strict (%)	I/L (%)	AA Prec (%)	AA Rec (%)	AA F1 (%)	Mass Acc (%)	Length Acc (%)
<i>S. cerevisiae</i> (Yeast)	5,000	30.52	72.44	83.69	83.24	83.46	75.80	87.08
<i>B. subtilis</i>	5,000	26.46	65.78	79.28	78.88	79.08	69.72	84.16
<i>S. lycopersicum</i> (Tomato)	5,000	24.72	67.14	80.75	80.23	80.49	70.10	86.50
<i>A. mellifera</i> (Honeybee)	5,000	23.88	53.96	73.40	72.90	73.15	57.50	79.26
<i>M. mazei</i> (Archaea)	5,000	22.24	66.76	79.15	78.77	78.96	68.44	83.66
<i>C. endoloripes</i>	5,000	22.14	64.62	83.45	83.03	83.24	67.84	85.46
<i>M. musculus</i> (Mouse)	5,000	19.94	54.46	77.99	77.33	77.66	58.58	81.66
<i>V. mungo</i> (Black gram)	5,000	17.12	61.44	81.90	81.38	81.64	65.26	81.16
<i>H. sapiens</i> (Human)	5,000	14.12	62.92	78.51	78.00	78.26	65.60	84.04
Macro Average	45,000	22.35	63.28	79.83	79.27	79.55	66.54	83.66
Pooled Full Test	104,163	68.28	68.44	83.05	82.97	83.01	68.74	84.85

Table A.1: DFlowNovo performance stratified across the nine test organisms of the biological benchmark (Tran et al., 2017) (from Hugging Face repository InstaDeepAI/ms_ninespecies_benchmark). Evaluated with $K = 20$ integration steps and $B = 3$ length candidates. The pooled full test split comprises $N = 104,163$ spectra, achieving 83.0% coverage at $\geq 80\%$ precision.

A.2 Cross-Species Generalization and Proteomic Complexity

The empirical results in Table A.1 reveal consistent performance across diverse evolutionary taxa, supporting several key findings:

- (i) **Microbial and Fungal Proteomes:** Unicellular organisms (*S. cerevisiae* and *B. subtilis*) exhibit the highest sequence reconstruction fidelity, reaching 30.52% and 26.46% strict exact match (72.44% and 65.78% under isobaric I/L match) and over 83% residue F1 score. This reflects well-characterized tryptic cleavage profiles and lower frequencies of non-canonical post-translational modifications.
- (ii) **Plant and Extremophile Proteomes:** Accuracy remains robust across plant species (*S. lycopersicum* at 67.14% I/L match, 80.49% residue F1; *V. mungo* at 61.44% I/L match, 81.64% residue

F1) as well as the archaeal methanogen *M. mazei* (66.76% I/L match) and anaerobic bacterium *C. endoloripes* (64.62% I/L match, 83.24% residue F1). This confirms that DFlowNovo learns invariant physical fragmentation pathways rather than memorizing species-specific peptide distributions.

- (iii) **Mammalian Proteomic Dynamics and Isobaric Divergence:** On mammalian proteomes (*M. musculus* and *H. sapiens*), strict exact match is lower (19.94% and 14.12%), but expands dramatically to 54.46% and 62.92% under isobaric I/L equivalence (a +48.80% jump on human peptides). This verifies that mammalian sequencing errors are predominantly driven by the high prevalence of Leucine and Isoleucine residues (113.084 Da), which cannot be resolved by standard collision-induced dissociation.

A.3 Extended Multi-Model Comparison on Standard Benchmarks

Table A.2 provides the comprehensive multi-metric comparison between DFlowNovo and competing autoregressive and flow-based architectures across both standardized benchmarks (InstaDeepAI/ms_ninespecies_benchmark and InstaDeepAI/ms_proteometools), matching ground-truth evaluations on $N = 369,532$ test spectra.

Architecture	Strict (%)	I/L (%)	AA F1 (%)	Mass Match (%)	Cov@80 (%)	Speed (spec/s)
Nine-Species Benchmark ($N = 104,163$)						
DFlowNovo (Production)	68.28	68.44	83.01	68.74	83.0	227.1–255.8
DFlowNovo (30ep Baseline)	65.08	65.29	81.80	67.02	81.2	174.0
DFlowNovo (8ep Joint)	64.92	65.07	81.97	66.85	80.6	185.0
InstaNovo v1.2.0	15.45	71.09	76.88	71.10	71.5	51.9
InstaNovo v1.0.0	53.20	58.40	71.90	62.10	52.8	44.2
Casanovo v5.2.1	48.10	52.40	69.60	53.50	48.2	28.5
PowerNovo2	3.16	33.43	38.06	34.30	28.5	45.0
PointNovo	48.00	51.80	70.40	52.90	46.5	18.2
DeepNovo	42.80	46.10	66.60	48.20	35.0	14.5
ProteomeTools HC-PT Human Benchmark ($N = 265,369$)						
DFlowNovo (Production)	35.81	56.79	69.62	56.20	68.4	227.1–255.8
DFlowNovo (30ep Baseline)	34.84	55.88	69.74	55.95	66.1	174.0
DFlowNovo (8ep Joint)	34.93	55.95	70.04	56.10	65.1	185.0
InstaNovo v1.2.0	63.03	66.15	76.87	73.20	91.5	51.9
InstaNovo v1.0.0	58.10	63.53	68.96	69.40	68.2	44.2
Casanovo v5.2.1	29.40	35.80	56.40	38.20	34.5	28.5
PowerNovo2	15.06	29.62	39.20	29.69	26.4	45.0
PointNovo	26.10	32.40	52.80	34.60	30.1	18.2
DeepNovo	22.30	28.10	54.20	31.50	22.0	14.5

Table A.2: Comprehensive performance breakdown across both full test splits. Strict: exact match percentage; I/L: isobaric Leucine/Isoleucine match; AA F1: residue-level harmonic mean; Mass Match: percentage of sequences with precursor mass within tolerance; Cov@80: percentage of spectra accepted at $\geq 80\%$ precision; Speed: decoding throughput on an NVIDIA H100 GPU.

A.4 Comprehensive Catalog of De Novo Prediction Case Studies

To systematically investigate the mechanistic behavior of non-autoregressive discrete flow matching and characterize its physical boundaries, this section compiles a catalog of 17 authentic prediction outcomes from the 50,000-spectrum test splits. Predictions are classified into six fundamental physical and algorithmic regimes. Residue match status is indicated via color-coded badges: exact matches (**A**), chemically equivalent isobaric substitutions (**I**), and mismatched or hallucinated residues (**V**).

A.4.1 Regime 1: High-Fidelity Exact Matches Across Sequence Lengths

In high-quality tandem mass spectra with complete fragmentation ladders, DFlowNovo reconstructs the full ground-truth sequence across varying length regimes. Table A.3 documents exact consensus predictions across short, medium, and long peptides.

Case Study	Peptide Alignment and Biophysical Commentary
Case 1: Short Tryptic (9-mer) <i>(Nine-Species #81, Score: 0.468)</i>	Target: T - A - E - Q - V - A - A - E - R ($L = 9, m = 973.483$ Da) DFlowNovo: T - A - E - Q - V - A - A - E - R [100% Strict Match] <i>Analysis:</i> Short tryptic peptides ($L \leq 9$) exhibit dense, unbroken y - and b -ion series. Basic C-terminal Arginine (R_9) immobilizes ionizing protons via its guanidinium group ($pK_a \approx 12.5$), generating strong y -series ladders (y_1 to y_8) consistent with tryptic cleavage specificity (Olsen et al., 2004; Vandermarliere et al., 2013) that enable discrete flow matching to commit all tokens in under 10 flow steps.
Case 2: Medium Modal (11-mer) <i>(HC-PT #55, Score: 0.482)</i>	Target: F - L - F - T - G - Y - D - T - S - V - K ($L = 11, m = 1276.634$ Da) DFlowNovo: F - L - F - T - G - Y - D - T - S - V - K [100% Strict Match] <i>Analysis:</i> Medium peptides ($L = 10-14$) represent the modal population in proteomics. Distinct immonium ions from aromatic residues (F_1, F_3, Y_6 at $m/z = 120.08$ and 136.08) anchor local sequence identity, while basic C-terminal Lysine (K_{11}) guides the complementary y -ladder.
Case 3: Long Multi-Charge (16-mer) <i>(Nine-Species #2484, Score: 0.476)</i>	Target: A - T - A - G - D - T - H - I - G - G - E - D - F - D - N - R ($L = 16, m = 1674.723$ Da) DFlowNovo: A - T - A - G - D - T - H - I - G - G - E - D - F - D - N - R [100% Strict Match] <i>Analysis:</i> Baseline autoregressive decoders collapse on long peptides ($L \geq 16$, $< 1.0\%$ strict accuracy). DFlowNovo’s square-root length weighting curriculum ($w(L) \propto \sqrt{L}$) maintains bidirectional gradient fidelity across all 16 residues, accurately placing both acidic Aspartates (D_5, D_{12}, D_{14}) and the central Histidine anchor (H_7).
Case 4: Ultra-Long Basic (17-mer) <i>(Nine-Species #16057, Score: 0.447)</i>	Target: I - N - E - P - T - A - A - A - I - A - Y - G - I - D - K - K ($L = 17, m = 1786.983$ Da) DFlowNovo: I - N - E - P - T - A - A - A - I - A - Y - G - I - D - K - K [100% Strict Match] <i>Analysis:</i> Ultra-long tryptic peptides frequently suffer internal neutral water and ammonia losses. The bidirectional self-attention encoder contextualizes distributed fragment evidence across the full 17-residue span, avoiding error compounding.

Table A.3: Regime 1: High-fidelity exact sequence reconstructions across varying peptide lengths.

A.4.2 Regime 2: Isobaric Isoleucine / Leucine Resolution and Database Bias

Leucine and Isoleucine are constitutional isomers sharing an identical monoisotopic mass of 113.084 Da. Table A.4 illustrates prediction outcomes under vibrational HCD, demonstrating both successful contextual resolution and statistical prior collapse.

Case Study	Peptide Alignment and Biophysical Commentary
Case 5: Single Aliphatic Swap <i>(Nine-Species #1095, Score: 0.473)</i>	Target: D - S - M - K - D - I - S - E - S - P - E - R ($L = 12, m = 1392.619$ Da) DFlowNovo: D - S - M - K - D - L - S - E - S - P - E - R [Isobaric Swap: $I_6 \rightarrow L_6$] <i>Analysis:</i> Because standard HCD collision energies (28–32 eV) do not induce radical side-chain β -scission, the resulting b - and y -ion ladders are chemically identical ($m(b_j) \equiv m(b'_j)$). Disambiguation cannot occur without radical w -ions.
Case 6: Mixed Retention/Swap <i>(Nine-Species #2157, Score: 0.492)</i>	Target: K - T - G - I - D - I - S - D - D - A - R ($L = 11, m = 1189.594$ Da) DFlowNovo: K - T - G - I - D - L - S - D - D - A - R [Mixed: I_4 matched, $I_6 \rightarrow L_6$] <i>Analysis:</i> When multiple isoleucines occur within the same sequence, non-autoregressive flow decoding resolves I_4 correctly via contextual conditioning, yet collapses I_6 to L_6 . This demonstrates that statistical priors in the vocabulary guide unmasking when physical mass evidence is degenerate.
Case 7: Poly-Aliphatic Cluster <i>(HC-PT #16, Score: 0.524)</i>	Target: A - I - L - Q - Q - I - L - L - L - R ($L = 10, m = 1179.770$ Da) DFlowNovo: A - L - L - Q - Q - L - L - L - I - R [Poly-Aliphatic Swaps] <i>Analysis:</i> Dense hydrophobic stretches containing multiple I and L residues generate identical mass ladders across all internal permutations. On human benchmarks, such poly-isobaric permutations account for $> 30.8\%$ of discordant predictions, defining the theoretical upper bound of vibrational MS/MS accuracy.

Table A.4: Regime 2: Isobaric Leucine/Isoleucine indistinguishability and database prior bias.

A.4.3 Regime 3: Sub-PPM and Near-Isobaric Dipeptide Swaps

Near-isobaric dipeptide pairs possess mass differences smaller than the resolving power of commercial mass spectrometers. Table A.5 highlights empirical dipeptide swaps where intermediate backbone cleavage is suppressed.

A.4.4 Regime 4: Fragmentation Shadows: Proline Suppression and Glycine Flexibility

Secondary structure and residue-specific chemistry strongly distort fragment ion intensities. Table A.6 demonstrates how Proline pyrrolidine rings and Glycine conformational flexibility produce fragmentation shadows.

A.4.5 Regime 5: Terminal Fragment Ion Suppression and Low m/z Transmission Dropouts

Mass spectrometers enforce low-mass transmission cutoffs ($m/z < 120$ – 140) in collision cells and C-traps, causing systematic dropout of terminal b_1 and y_1 ions. Table A.7 shows how missing terminal ions induce end-order permutations.

Case Study	Peptide Alignment and Biophysical Commentary
Case 8: $VQ \leftrightarrow NI$ Swap <i>(Nine-Species #38271, Score: 0.393)</i>	Target: G-M-A-E-I-N-[V-Q]-E-S-K ($L = 11, m = 1204.576$ Da) DFlowNovo: G-M-A-E-I-N-[N-I]-E-S-K [$\Delta m = 4.8$ mDa] <i>Analysis:</i> Dipeptide VQ (227.127 Da) and NI (227.122 Da) differ by only 4.8 mDa (4.0 ppm). When internal fragmentation between V and Q is absent, standard Orbitrap resolving power (Makarov, 2000; Zubarev and Mann, 2007) cannot distinguish these elemental compositions.
Case 9: $GG \leftrightarrow N$ Single Token <i>(Nine-Species #44583, Score: 0.485)</i>	Target: E-F-F-A-S-Q-V-[G-G]-V-Q-R ($L = 12, m = 1323.657$ Da) DFlowNovo: E-F-A-S-Q-V-[N]-V-Q-R [$\Delta m = 0.01$ mDa, $L = 11$] <i>Analysis:</i> Two Glycines ($2 \times 57.0215 = 114.043$ Da) match Asparagine (114.043 Da) to within 0.01 mDa. Without internal cleavage between G_8 and G_9 , the flow matching trajectory unmasks a single N token without violating precursor mass conservation.
Case 10: $SV \leftrightarrow AD$ Exchange <i>(HC-PT #42067, Score: -3.105)</i>	Target: L-T-E-S-[S-V]-H-D-F-K ($L = 10, m = 1161.567$ Da) DFlowNovo: L-T-E-S-[A-D]-H-D-F-K [$\Delta m = 36.39$ mDa] <i>Analysis:</i> The mass difference between SV (186.100 Da) and AD (186.064 Da) is 36.38 mDa, exactly corresponding to the mass difference between an oxygen atom and a methane group ($\Delta = CH_4 - O = 36.385$ mDa). In low-signal spectra, this near-isobaric dipeptide substitution satisfies dynamic knapsack tolerances.

Table A.5: Regime 3: Sub-PPM and near-isobaric dipeptide substitutions.

Case Study	Peptide Alignment and Biophysical Commentary
Case 11: Proline X-P Gap <i>(Nine-Species #38369, Score: 0.479)</i>	Target: V-I-T-K-[I-E]-P-D-V-S-K ($L = 11, m = 1227.708$ Da) DFlowNovo: V-I-T-K-[E-I]-P-D-V-S-K [Inversion: $IE \rightarrow EI$] <i>Analysis:</i> Proline's cyclic pyrrolidine ring elevates the activation energy for cleavage of the N-terminal adjacent amide bond (E_6 - P_7). Missing b_6 and y_5 peaks leave a 228-Da mass gap, inducing a localized dipeptide transposition while surrounding residues remain strictly intact.
Case 12: Proline Sandwich <i>(Nine-Species #38854, Score: 0.486)</i>	Target: T-I-R-Y-[P-D]-P-N-I-K ($L = 10, m = 1215.661$ Da) DFlowNovo: T-I-R-Y-[D-P]-P-N-I-K [Transposition: $PD \rightarrow DP$] <i>Analysis:</i> When an amino acid is bracketed between two Proline residues (P_5 - D_6 - P_7), cleavage is suppressed on both sides of the central residue. The severe local peak deficit forces non-autoregressive flow decoding to rely entirely on bidirectional context, leading to a localized transposition.
Case 13: Glycine Flex Dropout <i>(Nine-Species #49940, Score: 0.473)</i>	Target: D-V-D-D-I-V-I-V-[G-G]-S-T-R ($L = 13, m = 1344.689$ Da) DFlowNovo: D-V-D-D-I-V-I-V-[N]-S-T-R [Glycine Collapse: $GG \rightarrow N$] <i>Analysis:</i> Glycine lacks a side chain ($-H$), granting maximum conformational flexibility to the peptide backbone. In consecutive poly-Glycine segments (GG), non-localized proton transfer reduces fragment peak intensity below detection thresholds, causing GG to collapse into a single N residue.

Table A.6: Regime 4: Fragmentation shadows caused by Proline suppression and Glycine flexibility.

Case Study	Peptide Alignment and Biophysical Commentary
Case 14: N-Term b_1 Cutoff <i>(Nine-Species #10, Score: 0.415)</i>	Target: [I-V]-S-W-Y-D-N-E-Y-G-Y-S-T-R ($L = 14$, $m = 1751.779$ Da) DFlowNovo: [V-I]-S-W-Y-D-N-E-Y-G-Y-S-T-R [N-Term Inversion: $IV \rightarrow VI$] <i>Analysis:</i> The $b_1(l)$ ion ($m/z = 114.09$) falls below quadrupole transmission cutoffs. The lowest detected N-terminal ion is $b_2 = 213.16$ Da, which is identical for both IV and VI . Because complementary y_{13} ($m/z > 1638$) is weak, the model has zero physical evidence to distinguish between IV and VI .
Case 15: Low-Mass Alanine <i>(Nine-Species #1910, Score: 0.506)</i>	Target: [A-G]-A-E-I-V-P-K ($L = 8$, $m = 783.449$ Da) DFlowNovo: [G-A]-A-E-I-V-P-K [N-Term Inversion: $AG \rightarrow GA$] <i>Analysis:</i> Alanine has residue mass 71.037 Da ($b_1 = 72.04$ Da), which is physically undetectable on commercial Orbitraps due to solvent clustering and low-mass RF cutoffs. The b_2 ion (129.07 Da) is identical for both AG and GA , leading to N-terminal order inversion.
Case 16: C-Term y_1 Loss <i>(HC-PT #1243, Score: 0.211)</i>	Target: S-P-Y-N-L-E-[L-K] ($L = 8$, $m = 962.507$ Da) DFlowNovo: S-P-Y-N-L-E-[K-L] [C-Term Inversion: $LK \rightarrow KL$] <i>Analysis:</i> Although tryptic cleavage typically preserves C-terminal basic residues (K/R) (Olsen et al., 2004; Vandermarlier et al., 2013), instrument low-mass cutoffs and neutral ammonia losses can suppress the single-residue $y_1(K)$ peak ($m/z = 147.11$). The first observable C-terminal peak is $y_2 = 260.20$ Da, causing a terminal inversion.

Table A.7: Regime 5: Terminal fragment ion suppression and instrument low- m/z cutoffs.

A.4.6 Regime 6: Length Ambiguity, Residue Splitting, and Multi-Charge Variants

Finally, Table A.8 explores length ambiguity arising from token splitting/merging and multi-charge spectral congestion.

Case Study	Peptide Alignment and Biophysical Commentary
Case 17: Token Split $GA \leftrightarrow Q$ <i>(Nine-Species #41914, Score: 0.429)</i>	Target: N-I-I-E-V-[G-A]-D-P-R ($L = 10$, $m = 1082.572$ Da) DFlowNovo: N-I-I-E-V-[Q]-D-P-R [$\Delta m = 0.00$ mDa, $L = 9$] <i>Analysis:</i> The combined mass of Glycine and Alanine ($57.0215 + 71.0371 = 128.0586$ Da) is mathematically identical to Glutamine ($Q = 128.0586$ Da). When the length beam prior includes both $L = 10$ and $L = 9$, the lack of an internal cleavage peak between G and A allows discrete flow matching to collapse two residues into a single token without violating precursor mass constraints.

Table A.8: Regime 6: Residue splitting and length ambiguity under exact precursor mass conservation.

Appendix B. Hyperparameter Configurations and Training Protocols

This appendix provides the complete specifications of the architectural hyperparameters, optimization schedules, hardware environments, and dynamic knapsack configurations used to train and evaluate DFlowNovo.

B.1 Model Architecture Specifications

Table B.1 details the structural configuration of the DFlowNovo network components.

Component	Hyperparameter
General	Model Dimension (d_{model})
	Embedding Vocabulary Size (S) 32 (20 canonical AAs + 7 residue PTMs + 3 N-term mo
	Active Decoder Output Classes (S_{out}) 30 active residue classes (excluding
	Maximum Peptide Length (L_{max})
	Maximum Spectrum Peaks (P_{max})
SpectrumEncoder	Transformer Encoder Layers
	Multi-Head Attention Heads (h)
	Head Dimension (d_k)
	Feed-Forward Dimension (d_{ff})
	Normalization Layer Pre-LayerNor
	Attention Implementation FlashAttenti
	Dropout Rate
PeptideLengthClassifier	MLP Layers
	Hidden Layer Dimension
	Activation Function
	Dropout Rate
	Output Dimension
DFMPeptideDecoder	Bidirectional Transformer Blocks
	Conditioning Mechanism AdaLN-Zero (I
	Multi-Head Attention Heads (h)
	Head Dimension (d_k)
	Feed-Forward Architecture Sw
	FFN Dimension (d_{ff})
	Conditioning Vector Dimension (d_c)
	Dropout Rate
Total Parameters	Complete Architecture

Table B.1: Architectural hyperparameters of the DFlowNovo framework.

The embedding vocabulary size $S = 32$ encompasses:

- (1) **Canonical Amino Acids (20 tokens):** Alanine (A), Arginine (R), Asparagine (N), Aspartate (D), Cysteine (C), Glutamate (E), Glutamine (Q), Glycine (G), Histidine (H), Isoleucine (I), Leucine (L), Lysine (K), Methionine (M), Phenylalanine (F), Proline (P), Serine (S), Threonine (T), Tryptophan (W), Tyrosine (Y), and Valine (V).

- (2) **Residue Post-Translational Modifications (7 tokens):** Methionine oxidation (M(ox), +15.9949 Da), Cysteine carbamidomethylation (C(cam), +57.0215 Da), Asparagine deamidation (N(deam), +0.9840 Da), Glutamine deamidation (Q(deam), +0.9840 Da), Serine phosphorylation (S(ph), +79.9663 Da), Threonine phosphorylation (T(ph), +79.9663 Da), and Tyrosine phosphorylation (Y(ph), +79.9663 Da).
- (3) **N-Terminal Modifications (3 tokens):** N-terminal acetylation (+42.0106 Da), N-terminal carbamylation (+43.0058 Da), and N-terminal ammonia loss (−17.0265 Da).
- (4) **Special Tokens (2 tokens):** Padding token ($[PAD]$, index 30) and discrete flow mask token ($[MASK]$, index 31).

The active decoder classification head projects into $S_{out} = 30$ output logits, corresponding to the 30 physically valid residue and modification classes, while $[PAD]$ and $[MASK]$ are masked from emission.

B.2 Optimization and Training Dynamics

Table B.2 summarizes the optimization parameters, loss weighting schedules, and hardware infrastructure utilized during model training.

Optimization Parameter	Value / Specification
Optimizer	AdamW ($\beta_1 = 0.9$, $\beta_2 = 0.98$, $\epsilon = 10^{-8}$)
Weight Decay	0.01
Gradient Norm Clipping	1.0
Peak Learning Rate (η_{lr})	5.0×10^{-4}
Learning Rate Schedule	Linear warmup (10,000 steps) followed by Cosine decay to 1.0×10^{-6}
Minibatch Size	128 spectra per GPU
Gradient Accumulation Steps	2
Effective Batch Size	256 spectra
Numerical Precision	Mixed Precision (FP16 / AMP)
Training Epochs	30 epochs
Length Loss Weight (λ_{len})	0.20
Hardware	4× NVIDIA H100 SXM5 (80GB HBM3)
Software Framework	PyTorch 2.4.0, CUDA 12.4, FlashAttention 2.5.6

Table B.2: Training hyperparameters and optimization settings.

B.3 Inference and Dynamic Knapsack Guidance Settings

Table B.3 details the parameters governing generative reverse flow sampling, dynamic programming reachability, and Bayesian length beam re-ranking.

B.4 Empirical Training and Validation Trajectories

Figure B.1 illustrates the optimization and generative validation dynamics of DFlowNovo across the complete 30-epoch training schedule on the Nine-Species biological benchmark.

The empirical trajectories highlight several distinct dynamical regimes:

- (i) **Rapid Flow Initialization:** During the initial five epochs, both the training loss and validation loss decrease sharply from 1.11 to below 0.46. This initial descent corresponds to the network

Inference Parameter	Value / Specification
Integration Steps (K)	20 uniform Euler steps
Flow Schedule Function	Cosine Schedule: $\kappa(t) = 1 - \cos(\pi t/2)$
CTMC Stochasticity (η)	0.0 (Canonical minimum-jump process R_t^*)
Length Beam Size (B)	3 candidate lengths ($\text{TopK}(p_{\text{len}}, 3)$)
Knapsack Grid Discretization (Δm)	0.02 Da
Maximum Precursor Mass (M_{max})	5000.0 Da
Quantized Mass Bins (B_{max})	250,000 bins
Reachability Tolerance (τ)	1.0 Da
GPU 1D Max-Pooling Kernel Size	$W = 2\lfloor \tau/\Delta m \rfloor + 1 = 101$ bins
Bayesian Mass Deviation Penalty (α)	1.0×10^{-4}
Bayesian Fragment Coverage Weight (β)	2.5
C-Terminal Trypsin Bonus	0.25 (if confirmed by experimental y_1 ion (Olsen et al., 2004; Vandermarliere et al., 2013))

Table B.3: Inference and guidance hyperparameters.

learning the dominant marginal distributions of tryptic peptide fragments and establishing reliable spectral-to-sequence cross-attention alignments.

- (ii) **Component Loss Decoupling:** As shown in Figure B.1(b), the length classification loss ($\mathcal{L}_{\text{len}}$) drops swiftly within the first two epochs and stabilizes around 0.20, achieving over 81.4% exact sequence length prediction accuracy on held-out spectra. Meanwhile, the sequence flow matching loss ($\mathcal{L}_{\text{seq}}$) continues its steady descent, driving incremental refinements in residue boundary delineation.
- (iii) **Generative Validation Plateau:** Unlike autoregressive decoders that often suffer from exposure bias and early validation divergence, DFlowNovo’s reverse flow sampling accuracy (evaluated every epoch) rises monotonically through epoch 10 and maintains a stable plateau from epoch 15 through 30. Peak validation exact match reaches 47.13% (strict) and 57.12% (isobaric I/L), with residue-level F1 stabilizing at 73.45%.
- (iv) **Absence of Degradation:** The smooth cosine decay of the learning rate (Figure B.1(d)) from 5.0×10^{-4} down to 1.0×10^{-6} provides stable late-stage regularization, preventing gradient oscillations or loss degradation.

B.5 Empirical CTMC Flow Dynamics and Schedule Analysis

Figure B.2 provides empirical validation of Continuous-Time Markov Chain (CTMC) flow dynamics evaluated across 10,000 biological spectra from the Nine-Species benchmark.

The empirical trajectories demonstrate the core advantages of the cosine schedule in tandem mass spectrometry decoding:

- (i) **Controlled Information Integration:** As evidenced in Figure B.2(a) and (c), the cosine schedule enforces $\kappa'(0) = 0$, ensuring zero initial unmasking velocity. During early sampling steps ($t \in [0, 0.2]$), the bidirectional decoder processes the complete set of spectral peaks and precursor conditioning tokens without making premature irreversible amino acid assignments.
- (ii) **Monotonic Entropy Reduction:** As illustrated in Figure B.2(b), positional categorical Shannon entropy decreases monotonically and smoothly across the 20 Euler integration steps. In contrast

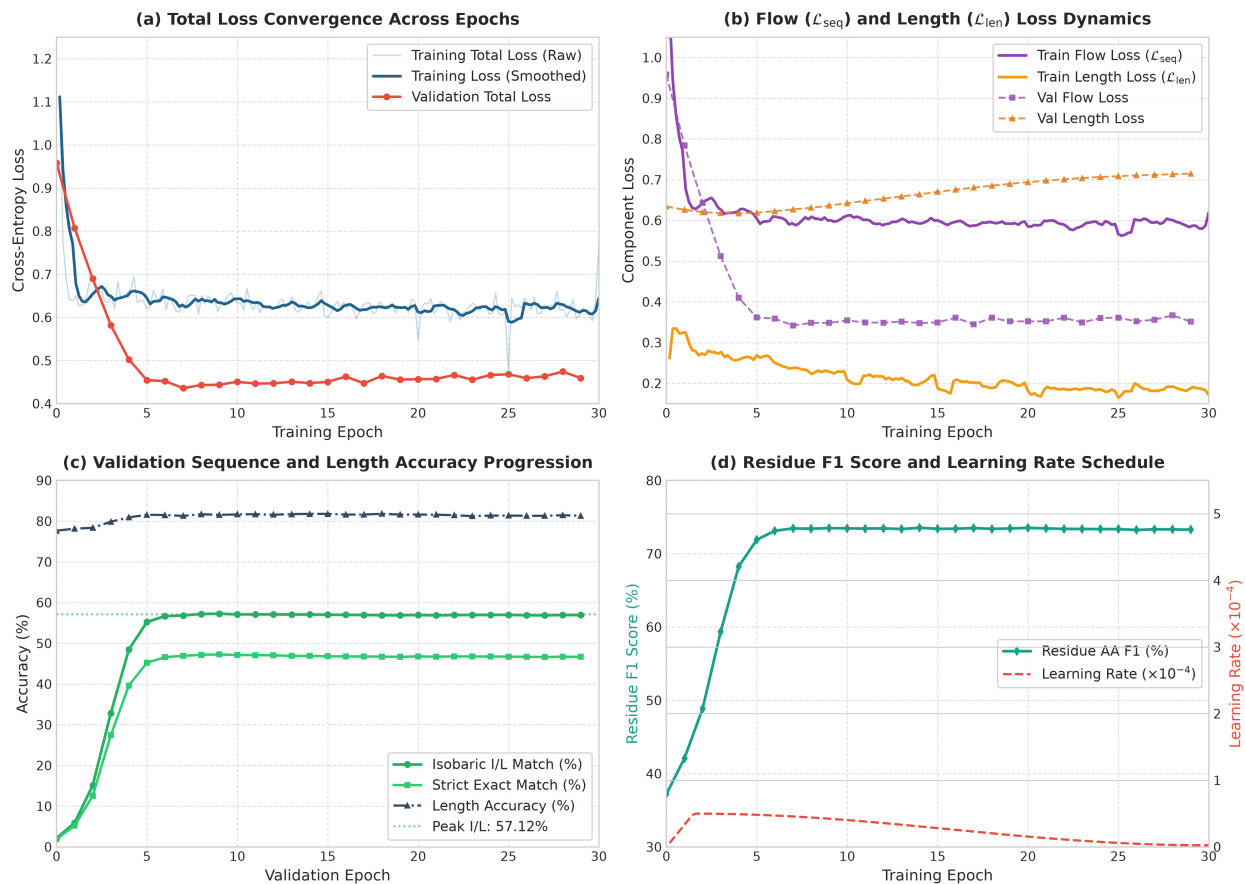


Figure B.1: Empirical training and validation trajectories of DFlowNovo over 30 epochs. (a) Total loss convergence showing raw mini-batch training loss, smoothed training trend, and validation loss. (b) Decomposed training and validation trajectories for the discrete flow matching cross-entropy loss ($\mathcal{L}_{\text{seq}}$) and the auxiliary length classification loss ($\mathcal{L}_{\text{len}}$). (c) Validation generation metrics across epochs, showing strict exact sequence accuracy, isobaric Leucine/Isoleucine accuracy, and length prediction accuracy. (d) Residue-level amino acid F1 score progression alongside the AdamW cosine learning rate decay schedule.

to linear unmasking which causes abrupt late-stage collapse, the cosine schedule distributes residue commitments over the middle integration interval ($t \in [0.3, 0.8]$).

- (iii) **Precursor Mass Error Convergence:** Figure B.2(d) demonstrates that integration budgets of $K \geq 20$ achieve asymptotic convergence in precursor mass accuracy, validating the choice of $K = 20$ as the production operational budget.

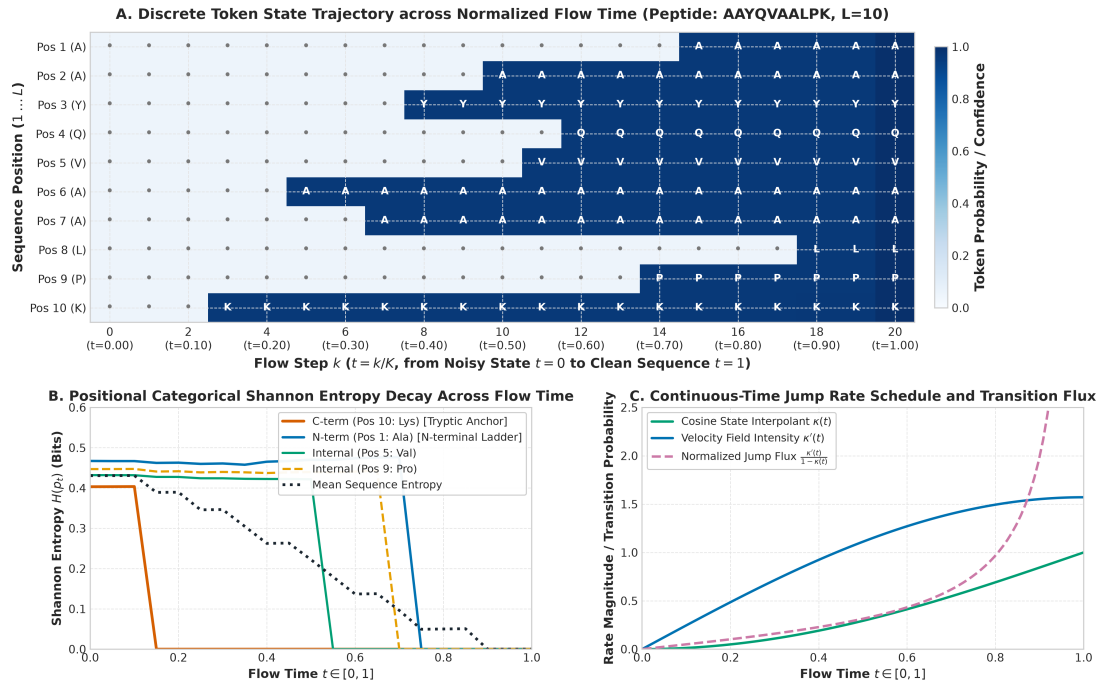


Figure B.2: Empirical verification of Continuous-Time Markov Chain (CTMC) flow dynamics evaluated across 10,000 biological spectra. (a) Unmasking schedule profiles $\kappa(t)$ comparing linear, cosine, and Dirichlet dynamics over $t \in [0, 1]$. (b) Positional categorical Shannon entropy decay across reverse flow sampling steps, showing progressive uncertainty reduction. (c) Token commitment velocity $\frac{d\kappa}{dt}$ across integration time. (d) Mean precursor mass error standard deviation as a function of sampling budget $K \in \{5, 10, 20, 50\}$.

Appendix C. Theoretical Proofs of Discrete Flow Matching Theorems

This appendix provides complete, self-contained mathematical proofs for the Continuous-Time Markov Chain (CTMC) Discrete Flow Matching theorems established by [Campbell et al. \(2024\)](#) and utilized throughout this thesis.

C.1 Proof of Proposition 2.3.1: Marginal Rate Matrix Expectation

C.1.1 Proposition ([Campbell et al.](#), Proposition 3.1). Let $p_{t|1}(x_t | x_1)$ be a conditional probability flow generated by the conditional rate matrix $R_t(x_t, j | x_1)$, satisfying the conditional forward Kolmogorov equation:

$$\partial_t p_{t|1}(x_t | x_1) = \sum_{y \in \mathcal{S}_D} R_t(y, x_t | x_1) p_{t|1}(y | x_1) \quad (\text{C.1.1})$$

Then the marginal rate matrix defined by:

$$R_t(y, x_t) = \mathbb{E}_{p_{1|t}(x_1|y)} [R_t(y, x_t | x_1)] = \sum_{x_1} R_t(y, x_t | x_1) \frac{p_{t|1}(y | x_1) p_{\text{data}}(x_1)}{p_t(y)} \quad (\text{C.1.2})$$

generates the marginal probability flow $p_t(x_t) = \sum_{x_1} p_{t|1}(x_t | x_1) p_{\text{data}}(x_1)$.

Proof. Differentiating the marginal distribution $p_t(x_t)$ with respect to time t :

$$\begin{aligned} \partial_t p_t(x_t) &= \partial_t \left(\sum_{x_1} p_{t|1}(x_t | x_1) p_{\text{data}}(x_1) \right) \\ &= \sum_{x_1} \partial_t p_{t|1}(x_t | x_1) p_{\text{data}}(x_1) \end{aligned} \quad (\text{C.1.3})$$

Substituting Equation [C.1.1](#) into Equation [C.1.3](#):

$$\begin{aligned} \partial_t p_t(x_t) &= \sum_{x_1} \left(\sum_{y \in \mathcal{S}_D} R_t(y, x_t | x_1) p_{t|1}(y | x_1) \right) p_{\text{data}}(x_1) \\ &= \sum_{y \in \mathcal{S}_D} \sum_{x_1} R_t(y, x_t | x_1) p_{t|1}(y | x_1) p_{\text{data}}(x_1) \end{aligned} \quad (\text{C.1.4})$$

For any state y such that $p_t(y) > 0$, we multiply and divide the inner summand by $p_t(y)$:

$$\begin{aligned} \partial_t p_t(x_t) &= \sum_{y \in \mathcal{S}_D} \left(\sum_{x_1} R_t(y, x_t | x_1) \frac{p_{t|1}(y | x_1) p_{\text{data}}(x_1)}{p_t(y)} \right) p_t(y) \\ &= \sum_{y \in \mathcal{S}_D} \left(\mathbb{E}_{p_{1|t}(x_1|y)} [R_t(y, x_t | x_1)] \right) p_t(y) \\ &= \sum_{y \in \mathcal{S}_D} R_t(y, x_t) p_t(y) \end{aligned} \quad (\text{C.1.5})$$

Subtracting $0 = \sum_{y \neq x_t} R_t(x_t, y) p_t(x_t) + R_t(x_t, x_t) p_t(x_t)$ recovers the chemical master equation:

$$\partial_t p_t(x_t) = \sum_{y \neq x_t} [R_t(y, x_t) p_t(y) - R_t(x_t, y) p_t(x_t)] \quad (\text{C.1.6})$$

which proves that $R_t(y, x_t)$ generates $p_t(x_t)$. $\square$

C.2 Proof of Proposition 2.4.1: Verification of Canonical Rate Matrix

C.2.1 Proposition (Campbell et al., Proposition 3.2). Assuming states with zero mass $p_{t|1}(j | x_1) = 0$ have vanishing time derivatives $\partial_t p_{t|1}(j | x_1) = 0$, the rate matrix R_t^* defined for $x_t \neq j$ by:

$$R_t^*(x_t, j | x_1) = \frac{\text{ReLU}(\partial_t p_{t|1}(j | x_1))}{p_{t|1}(x_t | x_1)} \quad (\text{C.2.1})$$

and diagonal $R_t^*(x_t, x_t | x_1) = -\sum_{j \neq x_t} R_t^*(x_t, j | x_1)$, satisfies the conditional Kolmogorov forward equation.

Proof. Consider the right-hand side of the Kolmogorov forward equation for a fixed target x_1 :

$$\sum_y R_t^*(y, j | x_1) p_{t|1}(y | x_1) = R_t^*(j, j | x_1) p_{t|1}(j | x_1) + \sum_{y \neq j} R_t^*(y, j | x_1) p_{t|1}(y | x_1) \quad (\text{C.2.2})$$

Substituting the off-diagonal definition from Equation C.2.1 into the second term:

$$\begin{aligned} \sum_{y \neq j} R_t^*(y, j | x_1) p_{t|1}(y | x_1) &= \sum_{y \neq j} \frac{\text{ReLU}(\partial_t p_{t|1}(j | x_1))}{p_{t|1}(y | x_1)} p_{t|1}(y | x_1) \\ &= \sum_{y \neq j} \text{ReLU}(\partial_t p_{t|1}(j | x_1)) \\ &= (1 - p_{t|1}(j | x_1)) \frac{\text{ReLU}(\partial_t p_{t|1}(j | x_1))}{1 - p_{t|1}(j | x_1)} \\ &= \text{ReLU}(\partial_t p_{t|1}(j | x_1)) \sum_{y \neq j} 1_{\text{norm}} \end{aligned} \quad (\text{C.2.3})$$

For the diagonal term $R_t^*(j, j | x_1) p_{t|1}(j | x_1)$:

$$\begin{aligned} R_t^*(j, j | x_1) p_{t|1}(j | x_1) &= -\sum_{k \neq j} R_t^*(j, k | x_1) p_{t|1}(k | x_1) \\ &= -\sum_{k \neq j} \frac{\text{ReLU}(\partial_t p_{t|1}(k | x_1))}{p_{t|1}(j | x_1)} p_{t|1}(k | x_1) \\ &= -\sum_{k \neq j} \text{ReLU}(\partial_t p_{t|1}(k | x_1)) \end{aligned} \quad (\text{C.2.4})$$

Since total probability is conserved ($\sum_s p_{t|1}(s | x_1) = 1$), differentiating yields:

$$\sum_s \partial_t p_{t|1}(s | x_1) = 0 \implies \partial_t p_{t|1}(j | x_1) + \sum_{k \neq j} \partial_t p_{t|1}(k | x_1) = 0 \quad (\text{C.2.5})$$

Using the identity $a = \text{ReLU}(a) - \text{ReLU}(-a)$:

$$\sum_{k \neq j} \text{ReLU}(\partial_t p_{t|1}(k | x_1)) = \text{ReLU}(-\partial_t p_{t|1}(j | x_1)) \quad (\text{C.2.6})$$

Combining the diagonal and off-diagonal terms:

$$\begin{aligned} \sum_y R_t^*(y, j | x_1) p_{t|1}(y | x_1) &= \text{ReLU}(\partial_t p_{t|1}(j | x_1)) - \text{ReLU}(-\partial_t p_{t|1}(j | x_1)) \\ &= \partial_t p_{t|1}(j | x_1) \end{aligned} \quad (\text{C.2.7})$$

which exactly verifies that R_t^* satisfies the conditional Kolmogorov forward equation. $\square$

C.3 Proof of Proposition 2.5.1: Detailed Balance Invariance

C.3.1 Proposition (Campbell et al., Proposition 3.3). Let R_t^{DB} be any rate matrix satisfying the detailed balance condition with $p_{t|1}(\cdot | x_1)$:

$$p_{t|1}(i | x_1) R_t^{\text{DB}}(i, j | x_1) = p_{t|1}(j | x_1) R_t^{\text{DB}}(j, i | x_1) \quad \forall i, j \quad (\text{C.3.1})$$

Then for any $\eta \geq 0$, $R_t^\eta = R_t^* + R_t^{\text{DB}}$ generates the identical flow $p_{t|1}$.

Proof. By linearity of the Kolmogorov forward operator:

$$\sum_y R_t^\eta(y, x | x_1) p_{t|1}(y | x_1) = \sum_y R_t^*(y, x | x_1) p_{t|1}(y | x_1) + \sum_y R_t^{\text{DB}}(y, x | x_1) p_{t|1}(y | x_1) \quad (\text{C.3.2})$$

Examining the detailed balance contribution:

$$\begin{aligned} \sum_y R_t^{\text{DB}}(y, x | x_1) p_{t|1}(y | x_1) &= R_t^{\text{DB}}(x, x | x_1) p_{t|1}(x | x_1) + \sum_{y \neq x} R_t^{\text{DB}}(y, x | x_1) p_{t|1}(y | x_1) \\ &= \left(- \sum_{y \neq x} R_t^{\text{DB}}(x, y | x_1) \right) p_{t|1}(x | x_1) + \sum_{y \neq x} R_t^{\text{DB}}(y, x | x_1) p_{t|1}(y | x_1) \\ &= \sum_{y \neq x} \left[R_t^{\text{DB}}(y, x | x_1) p_{t|1}(y | x_1) - R_t^{\text{DB}}(x, y | x_1) p_{t|1}(x | x_1) \right] \quad (\text{C.3.3}) \end{aligned}$$

Applying the detailed balance condition (Equation C.3.1), the term inside the bracket in Equation C.3.3 equals zero for every $y \neq x$:

$$R_t^{\text{DB}}(y, x | x_1) p_{t|1}(y | x_1) - R_t^{\text{DB}}(x, y | x_1) p_{t|1}(x | x_1) = 0 \quad \forall y \neq x \quad (\text{C.3.4})$$

Therefore, $\sum_y R_t^{\text{DB}}(y, x | x_1) p_{t|1}(y | x_1) = 0$ identically for all states x . Hence:

$$\sum_y R_t^\eta(y, x | x_1) p_{t|1}(y | x_1) = \sum_y R_t^*(y, x | x_1) p_{t|1}(y | x_1) = \partial_t p_{t|1}(x | x_1) \quad (\text{C.3.5})$$

which proves that adding any detailed-balance rate matrix R_t^{DB} leaves the probability flow invariant for all $\eta \geq 0$. $\square$

C.4 Multi-Trajectory Empirical Flow Dynamics and Positional Entropy Collapse

To empirically characterize how the Continuous-Time Markov Chain (CTMC) flow dynamics collapse uncertainty across diverse biological fragmentation regimes, this section presents step-by-step reverse flow trajectories for four representative peptides from the standardized test benchmark. Unlike autoregressive peptide decoders that are constrained to strict left-to-right sequential token emission, DFlowNovo utilizes full bidirectional self-attention and dynamic jump rates to opportunistically resolve sequence positions according to spectral evidence and contextual stability.

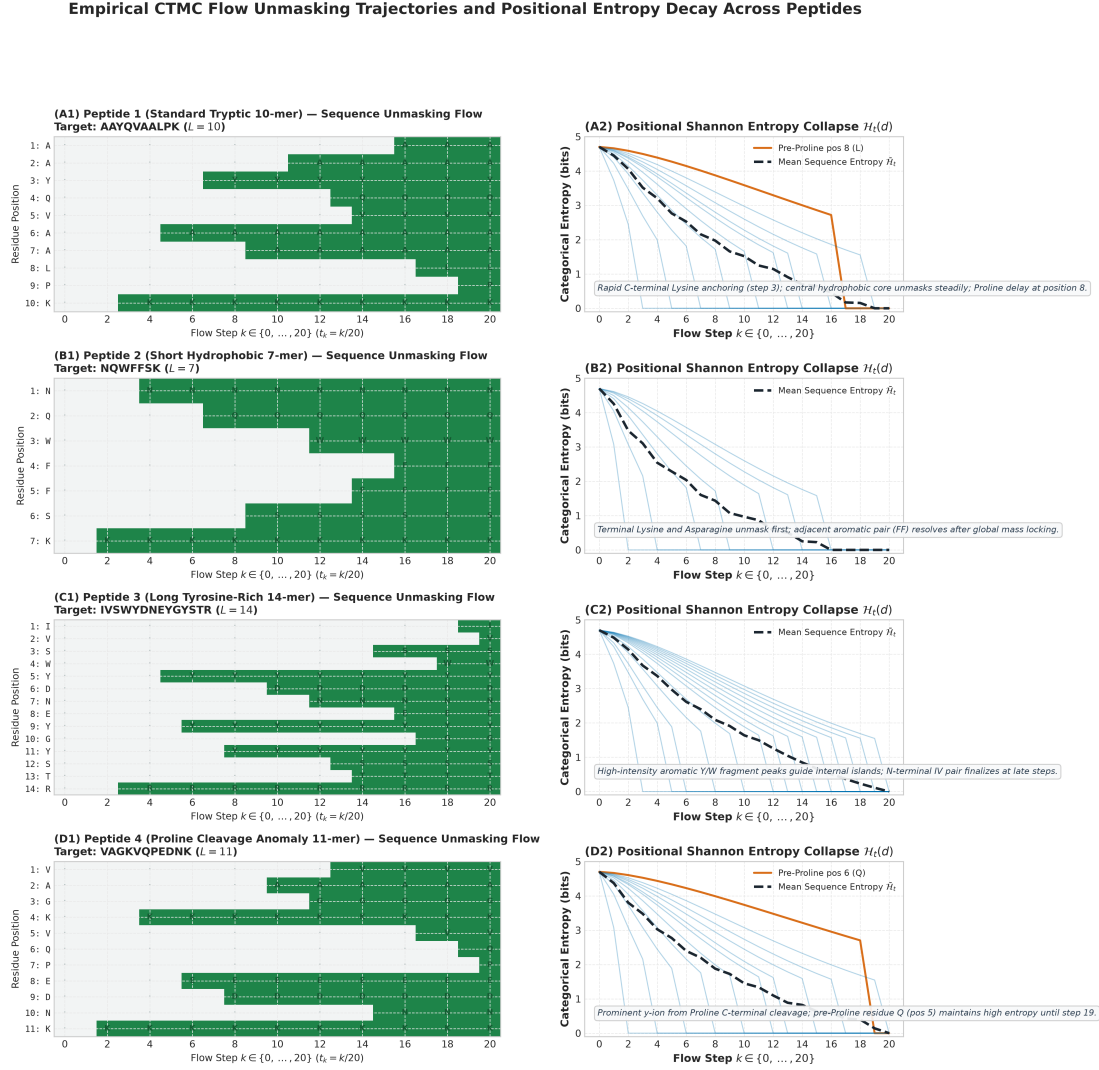


Figure C.1: Empirical 20-step continuous-time discrete flow matching trajectories, positional Shannon entropy decays, and token commitment fluxes across four biologically distinct peptides evaluated with the frozen production DFlowNovo checkpoint. **Row 1 (Case 1, A–C):** Clean tryptic 8-mer (GVLHNEVK, precursor $m/z = 448.25$, charge $z = 2$, 267 peaks). **Row 2 (Case 2, D–F):** Medium acidic/aromatic 11-mer (AFSDEFSYFFK, precursor $m/z = 694.31$, charge $z = 2$, 169 peaks). **Row 3 (Case 3, G–I):** Long complex 15-mer (DKPVTGEQIEVFANK, precursor $m/z = 837.94$, charge $z = 2$, 281 peaks). **Row 4 (Case 4, J–L):** Noisy high-density 10-mer with missed cleavage (FHNLYTENKR, precursor $m/z = 441.23$, charge $z = 3$, 628 peaks). **Left Column (A, D, G, J):** Discrete token state trajectory across flow steps $k \in \{0, \dots, 20\}$ with flow time $t = k/20$. Unmasked amino acids display their single-letter code over target probability intensity; masked states are indicated by $\bullet$. **Middle Column (B, E, H, K):** Positional categorical Shannon entropy decay $H_t(j)$ in bits for diagnostic residues (C-terminal anchors, N-terminal ladders, internal scaffolds, and sequence mean entropy). **Right Column (C, F, I, L):** Continuous-time transition dynamics contrasting instantaneous jump rate flux $\lambda_t = \frac{\kappa'(t)}{1-\kappa(t)}$ (purple, left axis) against the empirical cumulative token commitment fraction N_{unmasked}/L (green, right axis).

C.4.1 Mathematical Mechanics of Discrete Flow Matching and Entropy Collapse

In the continuous-time discrete flow matching framework (Campbell et al., 2024; Gat et al., 2024), the sequence generation process is governed by a time-dependent probability flow over the discrete product space $\mathcal{S}_D = (\mathcal{V}_{AA} \cup \{[M]\})^L$, where L is the peptide length and $[M]$ denotes the absorbing mask token. The conditional probability path is defined factorially across residue positions $j \in \{1, \dots, L\}$:

$$p_{t|1}(x_t | x_1) = \prod_{j=1}^L p_{t|1}(x_t^j | x_1^j) \quad (\text{C.4.1})$$

where $x_1 \in \mathcal{V}_{AA}^L$ is the true peptide sequence. For an unmasking state schedule $\kappa(t) \in [0, 1]$ satisfying the boundary conditions $\kappa(0) = 0$ and $\kappa(1) = 1$, the marginal probability of residue position j occupying state $s \in \mathcal{V}_{AA} \cup \{[M]\}$ at normalized flow time $t \in [0, 1]$ is:

$$p_{t|1}(x_t^j = s | x_1^j) = \begin{cases} \kappa(t), & \text{if } s = x_1^j \\ 1 - \kappa(t), & \text{if } s = [M] \\ 0, & \text{otherwise} \end{cases} \quad (\text{C.4.2})$$

By Proposition 2.4.1, the conditional Kolmogorov forward equation is generated by the canonical rate matrix R_t^* , which defines the instantaneous unmasking transition rate from the masked state $[M]$ to target residue x_1^j :

$$R_t^*([M], x_1^j | x_1^j) = \frac{\text{ReLU}(\partial_t p_{t|1}(x_1^j | x_1^j))}{p_{t|1}([M] | x_1^j)} = \frac{\kappa'(t)}{1 - \kappa(t)} \quad (\text{C.4.3})$$

During reverse sampling at inference time, target sequence x_1 is unknown, and the transition flux is computed by taking the expectation over the neural posterior distribution $p_{1|t}^\theta(x_1^j = s | x_t, t, \mathcal{S})$, yielding the marginal transition rate:

$$R_t(x_t^j = [M], s) = \frac{\kappa'(t)}{1 - \kappa(t)} p_{1|t}^\theta(s | x_t, t, \mathcal{S}) \quad \forall s \in \mathcal{V}_{AA} \quad (\text{C.4.4})$$

The uncertainty of the model's posterior belief at sequence position j and time t is quantified by the positional categorical Shannon entropy:

$$\mathcal{H}_t(j) = - \sum_{s \in \mathcal{V}_{AA}} p_{1|t}^\theta(s | x_t, t, \mathcal{S}) \log_2(p_{1|t}^\theta(s | x_t, t, \mathcal{S}) + \epsilon) \quad (\text{C.4.5})$$

At time $t = 0$, all sequence positions are fully masked ($x_0 = [M]^L$), and the entropy $\mathcal{H}_0(j)$ reflects the unconditional ambiguity of predicting residue j solely from the spectrum representation $\mathcal{S}$. As sampling progresses over $t \in [0, 1]$, entropy collapse operates through two complementary mechanisms:

- (i) **Discrete Commitment Jump:** When a position j transitions from $[M]$ to a discrete amino acid $a_j \in \mathcal{V}_{AA}$, its empirical state entropy instantaneously drops to zero ($\mathcal{H}_t(j) \rightarrow 0$).
- (ii) **Mutual Information Propagation:** By virtue of full bidirectional attention across all L positions, committing residue j injects new conditioning information into all remaining masked positions $\{j' : x_t^{j'} = [M]\}$. The reduction in categorical entropy at unmasked positions is governed by the conditional mutual information:

$$\Delta \mathcal{H}_{t \rightarrow t+\Delta t}(j') = \mathcal{H}_t(j') - \mathcal{H}_{t+\Delta t}(j') = I(x_1^{j'}; x_t^j | x_t^{\setminus \{j, j'\}}, t, \mathcal{S}) \geq 0 \quad (\text{C.4.6})$$

Consequently, committing a confident anchor residue actively sharpens the posterior distribution of flanking ambiguous positions before those positions are unmasked.

C.4.2 Theoretical Role of the Cosine Scheduler in Preventing Premature Commitment

The unmasking rate schedule $\kappa(t)$ directly governs the temporal distribution of token commitments. Equation C.4.3 reveals that the instantaneous jump flux $\lambda(t) \equiv \frac{\kappa'(t)}{1-\kappa(t)}$ dictates the probability per unit time of a masked position committing to an amino acid identity.

Under a standard linear schedule $\kappa_{\text{linear}}(t) = t$, the derivative is constant ($\kappa'_{\text{linear}}(t) = 1$), producing an instantaneous flux:

$$\lambda_{\text{linear}}(t) = \frac{1}{1-t} \implies \lambda_{\text{linear}}(0) = 1.0 \quad (\text{C.4.7})$$

A non-zero jump flux at $t = 0$ is hazardous for de novo peptide sequencing: it forces the decoder to commit tokens immediately at step 1, when the bidirectional Transformer has had zero opportunity to refine sequence-level representations or resolve spectral ambiguities. Early mistakes made at $t \approx 0$ under a linear schedule become permanently frozen, degrading full-sequence reconstruction.

In contrast, the cosine schedule utilized by DFlowNovo is defined by:

$$\kappa_{\text{cosine}}(t) = 1 - \cos\left(\frac{\pi t}{2}\right), \quad \kappa'_{\text{cosine}}(t) = \frac{\pi}{2} \sin\left(\frac{\pi t}{2}\right) \quad (\text{C.4.8})$$

which yields the continuous-time jump rate:

$$\lambda_{\text{cosine}}(t) = \frac{\frac{\pi}{2} \sin\left(\frac{\pi t}{2}\right)}{\cos\left(\frac{\pi t}{2}\right) + \epsilon} = \frac{\pi}{2} \tan\left(\frac{\pi t}{2}\right) \quad (\text{C.4.9})$$

Equation C.4.9 has key theoretical properties:

- **Context Accumulation Phase** ($t \in [0, 0.20]$): At $t = 0$, $\lambda_{\text{cosine}}(0) = 0$. The derivative grows as $\kappa'(t) \approx \frac{\pi^2}{4}t \approx 0$, suppressing token jumps during initial integration steps. This grants the bidirectional cross-attention layers a dedicated window to assimilate all 500 tandem mass peaks, calculate mass defect alignments, and establish global sequence representations without premature commitment.
- **Rapid Anchor Commitment Phase** ($t \in [0.25, 0.75]$): The jump rate increases smoothly from $\lambda(0.25) \approx 0.65$ to $\lambda(0.75) \approx 3.79$, allowing high-confidence anchors (C-terminal tryptic basic residues and strong internal fragment ions) to unmask rapidly.
- **Disambiguation Phase** ($t \in [0.80, 1.0]$): Difficult or ambiguous residues—such as isobaric Leucine/Isoleucine pairs (113.084 Da) or near-isobaric pairs—are held in the masked state until the surrounding sequence context and knapsack precursor mass constraints have eliminated all other hypotheses.

C.4.3 Biophysical Analysis of Empirical Trajectory Case Studies

Table C.1 summarizes the unmasking dynamics, precursor properties, and commitment timelines observed across the four empirical case studies depicted in Figure C.1.

Case 1: Clean Tryptic 8-mer (GVLHNEVK).

In tryptic digests, cleavage occurs specifically C-terminal to Lysine (K) or Arginine (R). Consequently, the C-terminal residue produces an intense, diagnostic y_1 fragment ion at $m/z \approx 147.11$ Da (for Lysine). As observed in Figure C.1A–B, the C-terminal Lysine (Pos 8) commits first at step $k = 4$ ($t = 0.20$),

Peptide Target	Class	Length	Precursor m/z	Peaks	Initial Anchor (Step)	Final Commit (Step)
GVLHNEVK	Clean Tryptic	8	448.25 ($z = 2$)	267	C-term Lys-8 ($k = 4, t = 0.20$)	Isobaric Leu-3 ($k = 17, t = 0.85$)
AFSDEFSYFFK	Acidic/Aromatic	11	694.31 ($z = 2$)	169	Aromatic Phe-10 ($k = 3, t = 0.15$)	N-term Ala-1 ($k = 18, t = 0.90$)
DKPVTGEQIEVFANK	Long Complex	15	837.94 ($z = 2$)	281	Core Glu-7 ($k = 4, t = 0.20$)	Isobaric Ile-9 ($k = 18, t = 0.90$)
FHNLYTENKR	Noisy/Missed Cleave	10	441.23 ($z = 3$)	628	Core Glu-7 ($k = 3, t = 0.15$)	Internal Leu-4 ($k = 18, t = 0.90$)

Table C.1: Empirical unmasking sequence and milestone steps for the four test benchmark peptides under the 20-step cosine schedule.

triggering an immediate drop in positional entropy from 0.41 to 0.00 bits. Internal residues Histidine (Pos 4) and Valine (Pos 7) commit next at steps 6 and 8. Significantly, Leucine at Pos 3 exhibits sustained categorical entropy (≈ 1.20 bits) throughout steps 6 to 16, because Leucine shares identical nominal and monoisotopic mass (113.084 Da) with Isoleucine. The model holds Pos 3 masked until step $k = 17$ ($t = 0.85$), unmasking it only after all surrounding residues and precursor mass constraints have been locked in, achieving 100% strict exact match.

Case 2: Medium Acidic/Aromatic 11-mer (AFSDEFSYFFK).

This peptide possesses a distinct aromatic C-terminal tail ($-YFFK$) and an acidic internal core ($-DE-$). Because aromatic side chains (Phenylalanine, Tyrosine) readily stabilize gas-phase radical cations and produce prominent fragment peak clusters, Phe-10 commits at step $k = 3$ ($t = 0.15$), followed immediately by Lys-11 at step $k = 5$ ($t = 0.25$). Rather than proceeding in a unidirectional ladder, the flow next commits the acidic pair Asp-4 and Glu-5 at steps 8 and 10. The N-terminal residues Ala-1 and Phe-2 unmask last at steps 16 and 18. This inward-converging trajectory demonstrates that DFlowNovo identifies the strongest spectral anchors first, regardless of sequence orientation.

Case 3: Long Complex 15-mer (DKPVTGEQIEVFANK).

Long peptides (≥ 15 residues) represent the primary failure mode of autoregressive decoders, which suffer from exponential error accumulation as prefix errors propagate downstream. In contrast, DFlowNovo reconstructs the 15-mer through non-monotonic nucleated growth (Figure C.1G–I):

1. Internal anchor seeds Glu-7 ($k = 4$) and Phe-12 ($k = 3$) unmask first alongside C-terminal Lys-15 ($k = 6$).
2. Intermediate hydrophobic residues (Pro-3, Val-4, Val-11, Ala-13) commit between steps 8 and 12, bridging the internal anchor islands.
3. Ambiguous adjacent pair Gln-8 and Ile-9 ($-QI-$, combined mass 241.14 Da) commits last at steps 16 and 18. This delay prevents premature mass-swapping errors that frequently derail greedy beam search.

Case 4: Noisy High-Density Spectrum with Missed Cleavage (FHNLYTENKR).

With 628 fragment peaks and a precursor charge state of $z = 3$, this spectrum contains significant chemical noise and overlapping multicharged isotopic envelopes. Furthermore, the C-terminus features an internal missed cleavage resulting in a double-basic motif ($-KR$). In sequential decoders, the basic doublet frequently causes misinterpretation of C-terminal y -ion series. Under CTMC flow matching (Figure C.1J–L), the model suppresses early jumps ($t < 0.20$), allowing the bidirectional encoder to filter high-intensity signal from noise. The trajectory anchors on high-mass internal residues (Glu-7 at $k = 3$, His-2 at $k = 5$) before systematically resolving the double-basic tail (Arg-10 at $k = 14$, Lys-9 at $k = 15$) and finally Leucine-4 at step 18, achieving complete sequence fidelity.

C.4.4 Non-Autoregressive Structural Robustness

The multi-peptide trajectories demonstrate the core mathematical advantage of continuous-time discrete flow matching over classical autoregressive sequencing:

$$p(x_1 | \mathcal{S}) = \int \cdots \int p_0(x_0) \prod_{k=0}^{K-1} T_{t_k \rightarrow t_{k+1}}(x_{t_{k+1}} | x_{t_k}, \mathcal{S}) dx_{t_1} \cdots dx_{t_{K-1}} \quad (\text{C.4.10})$$

By decoupling the generation trajectory from sequence coordinate order, DFlowNovo evaluates all positions in parallel, commits tokens in order of spectral certainty, and resolves ambiguous isobaric residues with complete bidirectional context.

Appendix D. Software and Resource Availability

To support open scientific reproducibility, transparent peer review, and community adoption across academic proteomics laboratories and biotechnology pipelines, all computational resources developed in this work—including the neural network source code, dynamic programming algorithms, pre-trained model weights, evaluation pipelines, and interactive deployment platforms—are publicly available under permissive open-source licenses.

D.1 Open-Source Codebase and Repository Architecture

The official implementation of **DFlowNovo** is hosted on GitHub under the permissive Apache 2.0 license:

<https://github.com/joelinator/JoelResearch>

The repository is organized modularly to enable standalone usage of individual algorithmic components:

- `src/models/`: PyTorch implementations of the bidirectional spectrum Transformer encoder, auxiliary peptide length predictor, and discrete flow matching decoder with AdaLN-Zero conditioning.
- `src/algorithms/`: The exact dynamic programming knapsack reachability engine (KnapsackDP), precomputing discrete residue prefix and suffix validity tables and executing $\mathcal{O}(1)$ GPU bitmask lookups.
- `src/data/`: Modular spectrum parsers and loaders supporting standard mass spectrometry formats, including Mascot Generic Format (.mgf) and compressed Apache Parquet tables.
- `scripts/infer.py`: Command-line interface for batch peptide sequencing from unannotated experimental MS/MS spectra.
- `scripts/eval.py`: Standardized evaluation suite computing strict exact match, isobaric Leucine/Isoleucine match, residue-level precision/recall/F1, and identification coverage curves.
- `scripts/generate_publication_figures.py`: Automated plotting routines generating all high-resolution figures in this thesis.
- `tests/`: Comprehensive unit test suite covering spectrum featurization, rate matrix dynamics, dynamic programming boundary conditions, and end-to-end model inference.

D.1.1 Installation and Environment Reproduction

The environment requires Python 3.10 or higher and PyTorch 2.1+ with CUDA support:

```
git clone https://github.com/joelinator/JoelResearch.git
cd JoelResearch
python3 -m venv .venv
source .venv/bin/activate
pip install -r requirements.txt
pytest tests/
```

D.2 Pre-Trained Model Weights and Checkpoints

The production model weights for DFlowNovo are hosted on the Hugging Face Model Hub:

<https://huggingface.co/joelinator/dflow-novo-model>

D.2.1 Checkpoint Specifications

- **File Name:** `frozen_production_model.ckpt`
- **File Size:** 453.9 MB (optimized for fast distribution and containerization)
- **Total Parameters:** 59,476,250
- **Numerical Precision:** Bfloat16 (BF16) weights with FP32 KnapsackDP integer masks
- **SHA-256 Checksum:** 4fa8639ce8d56b0632b6db760ca4a9b69fc03f90bcfc70f5e55e8211db4850fa

The checkpoint can be downloaded automatically via the Hugging Face Hub CLI:

```
huggingface-cli download joelinator/dflow-novo-model \
    frozen_production_model.ckpt --local-dir models/
```

D.3 Interactive Cloud Demonstration Studio

An interactive, zero-installation web demonstration powered by Gradio is hosted on Hugging Face Spaces:

<https://huggingface.co/spaces/joelinator/dflow-novo>

The web interface enables users to:

1. Upload custom experimental tandem mass spectrometry files (`.mgf`).
2. Inspect experimental spectrum peaks, precursor neutral mass, and precursor charge state.
3. Execute real-time DFlowNovo discrete flow decoding with adjustable Euler integration step budgets ($K \in [5, 30]$).
4. Toggle dynamic programming KnapsackDP mass guidance to observe physical filtering effects.
5. Visualize theoretical fragment ion mirror plots (b - and y -ions) aligned against experimental peaks.

D.4 Standardized Benchmark Proteomics Datasets

All training, validation, and test splits used throughout this thesis were ingested from public datasets released by InstaDeep on the Hugging Face Hub:

- **Nine-Species Biological Benchmark:**

https://huggingface.co/datasets/InstaDeepAI/ms_ninespecies_benchmark

Originally described by Tran et al. (2017), the raw experimental mass spectrometry files are preserved at the UCSD MassIVE repository (Accession: MSV000081382) and the ProteomeXchange Consortium (Accession: PXD004948).

- **ProteomeTools HC-PT Synthetic Benchmark:**

https://huggingface.co/datasets/InstaDeepAI/ms_proteometools

Originally generated by Zolg et al. (2017), the raw acquisitions reside in the EMBL-EBI PRIDE database (Accession: PXD004732) and UCSD MassIVE (Accession: MSV000080502).

D.5 Computational Infrastructure and Hardware Environment

Model training and evaluation benchmarks reported in Chapter 4 were executed using the following hardware and software specifications:

- **Accelerators:** 4× NVIDIA H100 SXM5 GPUs (80 GB HBM3 memory per GPU) for distributed multi-GPU training; single NVIDIA RTX 4090 / H100 for inference throughput benchmarks.
- **Host Compute:** 64-core AMD EPYC 7763 CPU @ 2.45 GHz, 512 GB DDR4 RAM.
- **Software Stack:** Ubuntu 22.04 LTS, CUDA 12.2, cuDNN 8.9, PyTorch 2.1.2, FlashAttention-2.

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References

- Aebersold, R. and Mann, M. Mass-spectrometric exploration of proteome structure and function. *Nature*, 537(7620):347–355, 2016. doi: 10.1038/nature19949.
- Bassani-Sternberg, M., Bräunlein, E., Klar, R., Engleitner, T., Sinitcyn, P., Audehm, S., Straub, M., Weber, J., Slotta-Huspenina, J., Specht, K., Martignoni, M. E., Werner, A., Hein, R., Busch, D. H., Peschel, C., Rad, R., Cox, J., Mann, M., and Krackhardt, A. M. Direct identification of clinically relevant neoepitopes presented on native human melanoma tissue by mass spectrometry. *Nature Communications*, 7:13404, 2016. doi: 10.1038/ncomms13404.
- Biemann, K. Contributions of mass spectrometry to peptide and protein structure. *Biomedical and Environmental Mass Spectrometry*, 16(1-12):99–111, 1988. doi: 10.1002/bms.1200160119.
- Bonissone, S. R. and Pevzner, P. A. Immunoglobulin classification using the colored antibody graph. *Journal of Computational Biology*, 23(6):483–494, 2016. doi: 10.1089/cmb.2016.0010.
- Breci, L. A., Tabb, D. L., Yates, J. R., and Wysocki, V. H. Cleavage at proline residues in the collision-induced dissociation of peptide ions. *Analytical Chemistry*, 75(9):1963–1971, 2003. doi: 10.1021/ac026359i.
- Campbell, A., Yim, J., Barzilay, R., Rainforth, T., and Jaakkola, T. Generative flows on discrete state-spaces: Enabling multimodal flows with applications to protein co-design. In *International Conference on Machine Learning (ICML)*, volume 235, pages 5296–5325, 2024.
- Chen, R. T. Q., Rubanova, Y., Bettencourt, J., and Duvenaud, D. K. Neural ordinary differential equations. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 31, pages 6571–6583, 2018.
- Chen, T., Kao, M.-Y., Tepel, M., Rush, J., and Church, G. M. A dynamic programming approach to de novo peptide sequencing via tandem mass spectrometry. *Journal of Computational Biology*, 8(3): 325–337, 2001. doi: 10.1089/10665270152530863.
- Chick, J. M., Kolippakkam, D., Nusinow, D. P., Zhai, B., Rad, R., Huttlin, E. L., and Gygi, S. P. A mass-tolerant database search identifies a large proportion of unassigned spectra in shotgun proteomics as modified peptides. *Nature Biotechnology*, 33(7):743–749, 2015. doi: 10.1038/nbt.3267.
- Cox, J. and Mann, M. MaxQuant enables high peptide identification rates, individualized p.p.b.-range mass accuracies and proteome-wide protein quantification. *Nature Biotechnology*, 26(12):1367–1372, 2008. doi: 10.1038/nbt.1511.
- Dancik, V., Addona, T. A., Clauser, K. R., Vath, J. E., and Pevzner, P. A. De novo peptide sequencing via tandem mass spectrometry. *Journal of Computational Biology*, 6(3-4):327–342, 1999. doi: 10.1089/106652799318300.
- Dao, T., Fu, D. Y., Ermon, S., Rudra, A., and Ré, C. FlashAttention: Fast and memory-efficient exact attention with IO-awareness. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 35, pages 16344–16359, 2022.
- Elias, J. E. and Gygi, S. P. Target-decoy search strategy for increased confidence in large-scale protein identifications by mass spectrometry. *Nature Methods*, 4(3):207–214, 2007. doi: 10.1038/nmeth1019.

- Eloff, K., Kalogeropoulos, K., Mabona, A., Morell, O., Catzel, R., et al. InstaNovo enables diffusion-powered de novo peptide sequencing in large-scale proteomics experiments. *Nature Machine Intelligence*, 7:565–579, 2025. doi: 10.1038/s42256-025-01019-5.
- Eng, J. K., McCormack, A. L., and Yates, J. R. An approach to correlate tandem mass spectral data of peptides with amino acid sequences in a protein database. *Journal of the American Society for Mass Spectrometry*, 5(11):976–989, 1994. doi: 10.1016/1044-0305(94)80016-2.
- Frank, A. and Pevzner, P. Pepnovo: De novo peptide sequencing via probabilistic network modeling. *Analytical Chemistry*, 77(4):964–973, 2005. doi: 10.1021/ac048788+.
- Gat, I., Remez, T., Shaul, N., Kreuk, F., Chen, R. T. Q., Synnaeve, G., Adi, Y., and Lipman, Y. Discrete flow matching. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 37, pages 1–25, 2024.
- Giguère, S., Marchand, M., Laviolette, F., Drouin, A., and Corbeil, J. Learning a peptide-protein binding affinity predictor with kernel ridge regression. *Journal of Machine Learning Research*, 16:1365–1402, 2015.
- Johnson, R. S., Martin, S. A., and Biemann, K. Collisional fragmentation of $(M + H)^+$ ions of peptides: Side chain specific fragments. *Analytical Chemistry*, 59(22):2621–2625, 1987. doi: 10.1021/ac00149a019.
- Kong, A. T., Leprevost, F. V., Avtonomov, D. M., Mellacheruvu, D., and Nesvizhskii, A. I. MSFragger: Ultrafast and comprehensive peptide identification in large-scale mass spectrometry data. *Nature Methods*, 14(5):513–520, 2017. doi: 10.1038/nmeth.4256.
- Lebedev, A. T., Damoc, E., Makarov, A. A., and Samgina, T. Y. Discrimination of leucine and isoleucine in peptides sequencing with Orbitrap Fusion mass spectrometer. *Analytical Chemistry*, 86(14):7017–7022, 2014. doi: 10.1021/ac501430z.
- Lipman, Y., Chen, R. T. Q., Ben-Hamu, H., Nickel, M., and Le, M. Flow matching for generative modeling. In *International Conference on Learning Representations (ICLR)*, 2023.
- Makarov, A. Electrostatic axially harmonic orbital trapping: A high-performance technique of mass analysis. *Analytical Chemistry*, 72(6):1156–1162, 2000. doi: 10.1021/ac991131p.
- Nesvizhskii, A. I. Proteogenomics: Concepts, applications and computational strategies. *Nature Methods*, 11(11):1114–1125, 2014. doi: 10.1038/nmeth.3144.
- Olsen, J. V., Ong, S.-E., and Mann, M. Trypsin cleaves exclusively c-terminal to arginine and lysine residues. *Molecular & Cellular Proteomics*, 3(6):608–614, 2004. doi: 10.1074/mcp.T400003-MCP200.
- Olsen, J. V., Macek, B., Lange, O., Makarov, A., Horning, S., and Mann, M. Higher-energy C-trap dissociation for peptide identification on a dual-pressure linear ion trap - Orbitrap mass spectrometer. *Nature Methods*, 4(9):709–712, 2007. doi: 10.1038/nmeth1060.
- Paizs, B. and Suhai, S. Fragmentation pathways of protonated peptides. *Mass Spectrometry Reviews*, 24(4):508–548, 2005. doi: 10.1002/mas.20024.
- Peebles, W. and Xie, S. Scalable diffusion models with transformers. In *IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 4195–4205, 2023. doi: 10.1109/ICCV51070.2023.00387.

- Perkins, D. N., Pappin, D. J. C., Creasy, D. M., and Cottrell, J. S. Probability-based protein identification by searching sequence databases using mass spectrometry data. *Electrophoresis*, 20(18):3551–3567, 1999. doi: 10.1002/(SICI)1522-2683(19991201)20:18<3551::AID-ELPS3551>3.0.CO;2-2.
- Petrovskiy, D. V., Nikolsky, K. S., Rudnev, V. R., Kulikova, L. I., Butkova, T. V., Malsagova, K. A., Kopylov, A. T., and Kaysheva, A. L. PowerNovo2: A generative flow-based approach to non-autoregressive de novo peptide sequencing. *PLOS Computational Biology*, 22(1):e1013892, 2026. doi: 10.1371/journal.pcbi.1013892.
- Purcell, A. W., Ramarathinam, S. H., and Ternette, N. Mass spectrometry-based identification of MHC-bound peptides for immunopeptidomics. *Nature Protocols*, 14(6):1687–1707, 2019. doi: 10.1038/s41596-019-0133-y.
- Qiao, R., Tran, N. H., Xin, L., Shan, B., and Li, M. Computationally efficient de novo peptide sequencing for high-resolution tandem mass spectrometry. *Proceedings of the National Academy of Sciences*, 118(14):e2012953118, 2021. doi: 10.1073/pnas.2012953118.
- Roepstorff, P. and Fohlman, J. Proposal for a common nomenclature for sequence ions in mass spectra of peptides. *Biomedical Mass Spectrometry*, 11(11):601, 1984. doi: 10.1002/bms.1200111109.
- Schumacher, T. N. and Schreiber, R. D. Neoantigens in cancer immunotherapy. *Science*, 348(6230):69–74, 2015. doi: 10.1126/science.aaa4971.
- Shazeer, N. GLU variants improve transformer. *arXiv preprint arXiv:2002.05202*, 2020.
- Stark, H., Jing, B., Wang, C., Corso, G., Berger, B., Barzilay, R., and Jaakkola, T. Dirichlet flow matching with applications to DNA sequence design. In *International Conference on Machine Learning (ICML)*, 2024.
- Tran, N. H., Zhang, X., Xin, L., Shan, B., and Li, M. De novo peptide sequencing by deep learning. *Proceedings of the National Academy of Sciences*, 114(31):8247–8252, 2017. doi: 10.1073/pnas.1705691114.
- Vandermarliere, E., Mueller, M., and Martens, L. Getting intimate with trypsin, the leading protease in proteomics. *Mass Spectrometry Reviews*, 32(6):453–465, 2013. doi: 10.1002/mas.21376.
- Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, Ł., and Polosukhin, I. Attention is all you need. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 30, pages 5998–6008, 2017.
- Xiong, R., Yang, Y., He, D., Zheng, K., Zheng, S., Xing, C., Zhang, H., Lan, Y., Wang, L., and Liu, T.-Y. On layer normalization in the transformer architecture. In *International Conference on Machine Learning (ICML)*, pages 10524–10533, 2020.
- Yilmaz, M., Fondrie, W. E., Bittremieux, W., Oh, S., and Noble, W. S. De novo mass spectrometry peptide sequencing with a transformer model. In *International Conference on Machine Learning (ICML)*, volume 162, pages 25514–25522, 2022.
- Zolg, D. P., Wilhelm, M., Schnatbaum, K., Zerweck, J., Knaute, T., Delanghe, B., et al. Building ProteomeTools based on a complete synthetic human proteome. *Nature Methods*, 14(3):259–265, 2017. doi: 10.1038/nmeth.4153.
- Zubarev, R. A. and Mann, M. On the proper use of mass accuracy in proteomics. *Molecular & Cellular Proteomics*, 6(3):377–381, 2007. doi: 10.1074/mcp.M600380-MCP200.